

Part 4 The Global Monetary System

10

THE FOREIGN EXCHANGE MARKET

Learning Objectives

After reading this chapter, you will be able to:

- LO10-1** Describe the functions of the foreign exchange market.
- LO10-2** Understand what is meant by spot exchange rates.
- LO10-3** Recognize the role that forward exchange rates play in insuring against foreign exchange risk.
- LO10-4** Understand the different theories explaining how currency exchange rates are determined and their relative merits.
- LO10-5** Identify the merits of different approaches toward exchange rate forecasting.
- LO10-6** Compare and contrast the differences among translation, transaction, and economic exposure, and explain the implications for management practice.



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opening case

WHAT AILS TÜRKİYE?

A graph of the value of the Turkish lira against the U.S. dollar looks like a ski slope that declines steadily and then suddenly gets very steep. In early 2008, one Turkish lira was worth \$0.85 U.S. dollar (USD). Since then, the value of the lira has declined persistently, year in year out, with hardly a break, hitting an all-time low of \$0.053 USD in early March 2023. Put differently, the Turkish lira has lost almost 94 percent of its value against the U.S. dollar since 2008. Similar declines have occurred against other currencies. Against the euro, the lira has declined 91 percent. Against the British pound, it has fallen 90 percent. Although the lira has been declining against other currencies since 2008, the decline accelerated sharply in 2021 and 2022, pushing the country into an economic crisis.

The principle cause of the lira's fall has been high inflation. Between 2005 and 2020, inflation in Türkiye hovered between 5 and 10 percent, high by the standards of many nations, but not catastrophic. Meanwhile, the economy grew at a healthy pace, averaging 5.8 percent GDP growth per annum between 2002 and 2021. In 2021, however, the annual inflation rate accelerated to 22 percent. By late 2022, it had reached a 25-year high of 85.5 percent!

The surging inflation rate rapidly eroded the value of the lira against other currencies. Items imported from abroad increased sharply in price, the production costs of Turkish businesses spiked, while the real value of people's wages fell. Many skilled workers have responded by leaving Türkiye to look for employment elsewhere. The loss of such talent has the potential to cause long-term harm to the country's economy. The only silver lining to these storm clouds has been strong demand for Turkish exports, which are now cheaper when converted into foreign currencies.

While inflation became a worldwide problem in 2021–2022 due to the supply chain disruptions and shortages caused by the COVID pandemic, and to strong aggregate demand as governments responded to the pandemic by pumping money into their economies to keep recessionary forces at bay, the experience of Türkiye has been uniquely bad. The primary reason is that the Turkish economy has been running too hot, fueled by very low interest rates. Low interest rates stimulate demand by making credit cheaper, encouraging individuals and businesses to borrow more money. The resulting expansion of the money supply, however, can fuel inflation, particularly if demand is already strong, as it was in Türkiye.

The classic response to accelerating inflation is to increase interest rates, thereby tightening credit, reducing borrowing, contracting the money supply, and slowing down demand for goods and services. Such monetary tightening is designed to cool a hot economy and reduce inflation. The central bank of Türkiye has done the opposite: It has kept real interest rates very low. This has expanded the money supply, stimulated demand, and increased inflation, leading to a collapse in the value of the lira, thereby increasing import prices and further pumping up inflation. It is a very dangerous spiral that is sapping confidence in the Turkish economy. Foreign and local investors have been pulling their money out of Türkiye because of the lira's free fall, which puts further pressure on the lira and does long-term damage to the economy as business investment falls.

The low interest rate policy is largely due to President Recep Tayyip Erdogan's "unorthodox" understanding of monetary policy. Erdogan has been prime minister and then president of Türkiye since 2003. Historically, Erdogan has won elections by keeping interest rates low and flooding the economy with cheap money. There also seems to be an ideological aspect to his low interest rate policy. Erdogan has quoted Islam's aversion to usury, which views lending money at interest as sinful, as one of the reasons for the 2021 lowering of interest rates. He also seems to believe that low interest rates and strong demand will curb inflation—a view that has no support among economists.

In most nations, the central bank is independent of the government and would be insulated from political pressure. This is not the case in Türkiye, however, where the government can exercise direct control over the central bank and its policies. Making matters worse, Erdogan has dismissed multiple finance ministers and heads of the central bank who disagreed with and challenged his unconventional policies.

None of this is lost on the population of Türkiye. The popularity of Erdogan and his ruling AKP party plummeted in opinion polls, suggesting that he might lose power in elections held in June 2023. Erdogan's response was to pump even more money into the economy—raising the minimum wage and public-sector spending—while clamping down on his opponents, including jailing some on dubious charges. In the event, he won the election (albeit by a narrow margin) and remains in power.

Sources: C. Ostroff, "Why the Turkish Lira Is Falling Like a Rock," *The Wall Street Journal*, December 7, 2021; "The Lastest Desperate Attempt to Prop Up the Turkish Lira," *The Economist*, June 30, 2022; J. Askew, "Soaring Inflation and a Collapsing Currency: Why Is Turkey's Economy Such a Mess?" *euronews*, December 21, 2022.

foreign exchange market

A market for converting the currency of one country into that of another country.

Introduction

In this chapter, we shall take a close look at workings of the foreign exchange market. The **foreign exchange market** is a market for converting the currency of one country into that of

another country. An **exchange rate** is simply the rate at which one currency is converted into another—U.S. dollars into euros, euros into Türk lira, British pounds into Japanese yen, and so on. For example, Toyota uses the foreign exchange market to convert the dollars it earns from selling cars in the United States into Japanese yen. Without the foreign exchange market, international trade and international investment on the scale that we see today would be impossible; companies would have to resort to barter. The foreign exchange market is the lubricant that enables companies based in countries that use different currencies to trade with each other.

exchange rate

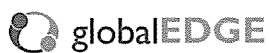
The rate at which one currency is converted into another.

We know from earlier chapters that international trade and investment have their risks. Some of these risks exist because future exchange rates cannot be perfectly predicted. The rate at which one currency is converted into another can change over time. For example, at the start of 2001, one U.S. dollar bought 1.065 euros, but by early 2014, one U.S. dollar bought only 0.74 euro. The dollar had fallen sharply in value against the euro. This made American goods cheaper in Europe, boosting export sales. At the same time, it made European goods more expensive in the United States, which hurt the sales and profits of European companies that sold goods and services to the United States. However, some of the pricing advantage enjoyed by U.S. companies disappeared between 2015 and 2022 as economic weakness in Europe and a stronger U.S. economy resulted in a fall in the value of the euro. By late 2022, one U.S. dollar bought 0.95 euro, meaning that American exports to the eurozone had become more expensive. Changes in currency values such as these often take managers by surprise, and if they have not hedged against the possible risk, sales and profits can be significantly affected.

One function of the foreign exchange market is to provide some insurance against the risks that arise from such volatile changes in exchange rates, commonly referred to as *foreign exchange risk*. Although the foreign exchange market offers some insurance against foreign exchange risk, it cannot provide complete insurance. International businesses frequently suffer losses (or gains) because of unpredicted changes in exchange rates. Currency fluctuations can make seemingly profitable trade and investment deals unprofitable, and vice versa.

This chapter has three main objectives. The first is to explain how the foreign exchange market works. The second is to examine the forces that determine exchange rates and to discuss the degree to which it is possible to predict future exchange rate movements. The third objective is to map the implications for international business of exchange rate movements. The opening case provides some insight into the second and third of these objectives. As described in the case, the rate of price inflation in a country can have a profound impact on the value of its currency—as it apparently has had in the case of the Turkish lira, where high inflation has resulted in a sharp depreciation in the value of the lira on the foreign exchange market. This has had a negative impact on businesses in Türkiye that rely on imports, since they have seen the cost of their inputs surge. On the other hand, businesses in Türkiye that export goods to other nations have seen strong demand since their prices, when converted into other currencies (such as euros and dollars), have fallen.

This chapter is the first of two that deals with the international monetary system and its relationship to international business. Chapter 11 explores the institutional structure of the international monetary system. The institutional structure is the context within which the foreign exchange market functions. As we shall see, changes in the institutional structure of the international monetary system can exert a profound influence on the development of foreign exchange markets.



globalEDGE™ Database of International Business Statistics

The “global money system” can have a significant effect on how companies operate globally. Often, companies have to deal with exchange rates, monetary systems, and the capital market on both country and regional levels. But the influences of countries on the regional and global money system are significant (i.e., countries set the tone for the parameters of the foreign exchange market and the international monetary system). The globalEDGE™ Database of International Business Statistics (DIBS) includes time-series

data beginning in the 1990s until today and covers more than 200 countries and more than 5,000 data variables. Countries, regions, and the world use these types of data points to drive the global money system, and everyone who is interested in better understanding the global capital market needs to know about them! Register free on globalEDGE™ to gain access to the DIBS database right now; students have free access to DIBS, and DIBS can be found at globaledge.msu.edu/tools-and-data/dibs.

We begin this chapter by looking at the functions and the form of the foreign exchange market. This includes distinguishing among spot exchanges, forward exchanges, and currency swaps. Then, we consider the factors that determine exchange rates. We also look at how foreign trade is conducted when a country's currency cannot be exchanged for other currencies—that is, when its currency is not convertible. The chapter closes with a discussion of these things in terms of their implications for business.

The Functions of the Foreign Exchange Market



LO10-1

Describe the functions of the foreign exchange market.

foreign exchange risk

The risk that changes in exchange rates will hurt the profitability of a business deal.

The foreign exchange market serves two main functions. The first is to convert the currency of one country into the currency of another. The second is to provide some insurance against **foreign exchange risk** or the adverse consequences of unpredictable changes in exchange rates.¹

CURRENCY CONVERSION

Each country has a currency in which the prices of goods and services are quoted. In the United States, it is the dollar (\$); in Great Britain, the pound (£); in France, Germany, and the other 17 members of the eurozone, it is the euro (€); in Japan, the yen (¥); and so on. In general, within the borders of a particular country, one must use the national currency. A U.S. tourist cannot walk into a store in Edinburgh, Scotland, and use U.S. dollars to buy a bottle of Scotch whisky. Dollars are not recognized as legal tender in Scotland; the tourist must use British pounds. Fortunately, the tourist can go to a bank, exchange dollars for pounds, and then buy the whisky.

When tourists change one currency into another, they are participating in the foreign exchange market. The exchange rate is the rate at which the market converts one currency into another. For example, an exchange rate of $\text{€}1 = \$1.20$ specifies that 1 euro buys 1.20 U.S. dollars. The exchange rate allows us to compare the relative prices of goods and services in different countries. A U.S. tourist wishing to buy a bottle of Scotch whisky in Edinburgh may find that they must pay £30 for the bottle, knowing that the same bottle costs \$35 in the United States. Is this a good deal? Imagine the current pound/dollar exchange rate is $\text{£}1.00 = \$1.25$ (i.e., 1 British pound buys \$1.25). Our intrepid tourist takes a calculator and converts £30 into dollars (the calculation is 30×1.25) and finds that the bottle of Scotch costs the equivalent of \$37.50. The tourist is surprised that a bottle of Scotch whisky could cost less in the United States than in Scotland despite shipping costs (alcohol is taxed heavily in Great Britain).

Tourists are minor participants in the foreign exchange market; companies engaged in international trade and investment are major ones. International businesses have four main uses of foreign exchange markets. First, the payments a company receives for its exports, the income it receives from foreign investments, or the income it receives from licensing agreements with foreign firms may be in foreign currencies. To use those funds in its home country, the company must convert them to its home country's currency. Consider the Scotch distillery that exports its whisky to the United States. The distillery is paid in dollars,

but because those dollars cannot be spent in Great Britain, they must be converted into British pounds. Similarly, Toyota sells its cars in the United States for dollars; it must convert the U.S. dollars it receives into Japanese yen to use the funds in Japan.

Second, international businesses use foreign exchange markets when they must pay a foreign company for its products or services in its country's currency. For example, Dell buys many of the components for its computers from Malaysian firms. The Malaysian companies must be paid in Malaysia's currency, the ringgit, so Dell must convert money from dollars into ringgit to pay them.

Third, international businesses also use foreign exchange markets when they have spare cash that they wish to invest for short terms in money markets. For example, consider a U.S. company that has \$10 million it wants to invest for three months. The best interest rate it can earn on these funds in the United States may be 2 percent. Investing in a South Korean money market account, however, may earn 6 percent. Thus, the company may change its \$10 million into Korean won and invest it in South Korea. Note, however, that the rate of return it earns on this investment depends not only on the Korean interest rate but also on the changes in the value of the Korean won against the dollar in the intervening period.

Currency speculation is the fourth use of foreign exchange markets. **Currency speculation** typically involves the short-term movement of funds from one currency to another in the hopes of profiting from shifts in exchange rates. Consider again a U.S. company with \$10 million to invest for three months. Suppose the company suspects that the U.S. dollar is overvalued against the Japanese yen. That is, the company expects the value of the dollar to depreciate (fall) against that of the yen. Imagine the current dollar/yen exchange rate is $\$1 = ¥120$. The company exchanges its \$10 million into yen, receiving ¥1.2 billion ($\$10 \text{ million} \times 120 = ¥1.2 \text{ billion}$). Over the next three months, the value of the dollar depreciates against the yen until $\$1 = ¥100$. Now the company exchanges its ¥1.2 billion back into dollars and finds that it has \$12 million. The company has made a \$2 million profit on currency speculation in three months on an initial investment of \$10 million! In general, however, companies should beware, for speculation by definition is a very risky business. The company cannot know for sure what will happen to exchange rates. While speculators may profit handsomely if their speculation about future currency movements turns out to be correct, they can also lose vast amounts of money if it turns out to be wrong.

A kind of speculation that has become more common in recent years is known as the **carry trade**. The carry trade involves borrowing in one currency where interest rates are low and then using the proceeds to invest in another currency where interest rates are high.² For example, if the interest rate on borrowings in Japan is 1 percent but the interest rate on deposits in U.S. banks is 6 percent, it can make sense to borrow in Japanese yen, convert the money into U.S. dollars, and deposit it in an American bank. The trader can make a 5 percent margin by doing so, minus the transaction costs associated with changing one currency into another. The speculative element of this trade is that its success is based on a belief that there will be no adverse movement in exchange rates (or interest rates for that matter) that will make the trade unprofitable. However, if the yen were to rapidly increase in value against the dollar, then more U.S. dollars would be required to repay the original loan, and the trade could fast become unprofitable. The dollar/yen carry trade was actually very significant during the mid-2000s, peaking at more than \$1 trillion in 2007, when some 30 percent of trade on the Tokyo foreign exchange market was related to the carry trade.³ This carry trade declined in importance during 2008–2009 because interest rate differentials were falling as U.S. rates came down, making the trade less profitable. In late 2016, the dollar/yen carry trade became profitable again as negative interest rates in Japan, coupled with rising interest rates in the United States, made it profitable to borrow in yen again and convert the money into U.S. dollars.⁴



Tourists exchanging currency in Istanbul, Turkey.

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currency speculation

Involves short-term movement of funds from one currency to another in hopes of profiting from shifts in exchange rates.

carry trade

A kind of speculation that involves borrowing in one currency where interest rates are low and then using the proceeds to invest in another currency where interest rates are high.

INSURING AGAINST FOREIGN EXCHANGE RISK



LO10-2

Understand what is meant by spot exchange rates.

A second function of the foreign exchange market is to provide insurance against foreign exchange risk, which is the possibility that unpredicted changes in future exchange rates will have adverse consequences for the firm. When a firm insures itself against foreign exchange risk, it is engaging in *hedging*. To explain how the market performs this function, we must first distinguish among spot exchange rates, forward exchange rates, and currency swaps.

Spot Exchange Rates

spot exchange rate

The exchange rate at which a foreign exchange dealer will convert one currency into another that particular day.

When two parties agree to exchange currency and execute the deal immediately, the transaction is referred to as a spot exchange. Exchange rates governing such “on the spot” trades are referred to as spot exchange rates. The **spot exchange rate** is the rate at which a foreign exchange dealer converts one currency into another currency on a particular day. Thus, when our U.S. tourist in Edinburgh goes to a bank to convert dollars into pounds, the exchange rate is the spot rate for that day.

Spot exchange rates are reported on a real-time basis on many financial websites. An exchange rate can be quoted in two ways: as the amount of foreign currency one U.S. dollar will buy or as the value of a dollar for one unit of foreign currency. Thus, on March 7, 2023 at 12:52 p.m., Eastern Standard Time, \$1 bought €0.95, and €1 bought \$1.06.

Spot rates change continually, often on a minute-by-minute basis (although the magnitude of changes over such short periods is usually small). The value of a currency is determined by the interaction between the demand and supply of that currency relative to the demand and supply of other currencies. For example, if lots of people want U.S. dollars and dollars are in short supply, and few people want British pounds and pounds are in plentiful supply, the spot exchange rate for converting dollars into pounds will change. The dollar is likely to appreciate against the pound (or the pound will depreciate against the dollar). Imagine the spot exchange rate is £1 = \$1.25 when the market opens. As the day progresses, dealers demand more dollars and fewer pounds. By the end of the day, the spot exchange rate might be £1 = \$1.23. Each pound now buys fewer dollars than at the start of the day. The dollar has appreciated, and the pound has depreciated.

Forward Exchange Rates



LO10-3

Recognize the role that forward exchange rates play in insuring against foreign exchange risk.

Changes in spot exchange rates can be problematic for an international business. For example, a U.S. company that imports high-end cameras from Japan knows that in 30 days it must pay yen to a Japanese supplier when a shipment arrives. The company will pay the Japanese supplier ¥200,000 for each camera, and the current dollar/yen spot exchange rate is \$1 = ¥120. At this rate, each camera costs the importer \$1,667 (i.e., $1,667 = 200,000/120$). The importer

knows it can sell the cameras the day they arrive for \$2,000 each, which yields a gross profit of \$333 on each (\$2,000 – \$1,667). However, the importer will not have the funds to pay the Japanese supplier until the cameras are sold. If, over the next 30 days, the dollar unexpectedly depreciates against the yen, say, to $\$1 = ¥95$, the importer will still have to pay the Japanese company ¥200,000 per camera but in dollar terms that would be equivalent to \$2,105 per camera, which is more than the importer can sell the cameras for. A depreciation in the value of the dollar against the yen from $\$1 = ¥120$ to $\$1 = ¥95$ would transform a profitable deal into an unprofitable one.

To *insure* or *hedge* against this risk, the U.S. importer might want to engage in a forward exchange. A **forward exchange** occurs when two parties agree to exchange currency and execute the deal at some specific date in the future. Exchange rates governing such future transactions are referred to as **forward exchange rates**. For most major currencies, forward exchange rates are quoted for 30 days, 90 days, and 180 days into the future. In some cases, it is possible to get forward exchange rates for several years into the future. Returning to our camera importer example, let us assume the 30-day forward exchange rate for converting dollars into yen is $\$1 = ¥110$. The importer enters into a 30-day forward exchange transaction with a foreign exchange dealer at this rate, so the importer is guaranteed that it will have to pay no more than \$1,818 for each camera ($1,818 = 200,000/110$). This guarantees a profit of \$182 per camera ($\$2,000 - \$1,818$). The importer also insures itself against the possibility that an unanticipated change in the dollar/yen exchange rate will turn a profitable deal into an unprofitable one.

In this example, the spot exchange rate ($\$1 = ¥120$) and the 30-day forward rate ($\$1 = ¥110$) differ. Such differences are normal; they reflect the expectations of the foreign exchange market about future currency movements. In our example, the fact that \$1 bought more yen with a spot exchange than with a 30-day forward exchange indicates foreign exchange dealers expected the dollar to depreciate against the yen in the next 30 days. When this occurs, we say the dollar is selling at a discount on the 30-day forward market (i.e., it is worth less than on the spot market). Of course, the opposite can also occur. If the 30-day forward exchange rate were $\$1 = ¥130$, for example, \$1 would buy more yen with a forward exchange than with a spot exchange. In such a case, we say the dollar is selling at a premium on the 30-day forward market. This reflects the foreign exchange dealers' expectations that the dollar will appreciate against the yen over the next 30 days.

In sum, when a firm enters into a forward exchange contract, it is taking out insurance against the possibility that future exchange rate movements will make a transaction unprofitable by the time that transaction has been executed. Although many firms routinely enter into forward exchange contracts to hedge their foreign exchange risk, sometimes this can work against the company. An example is given in the accompanying Management Focus, which explains how the hedging strategy adopted by the Brazilian regional jet manufacturer, Embraer, backfired.

Currency Swaps

The preceding discussion of spot and forward exchange rates might lead you to conclude that the option to buy forward is very important to companies engaged in international trade—and you would be right. According to the most recent data, forward instruments account for almost two-thirds of all foreign exchange transactions, while spot exchanges account for about one-third.⁵ However, the vast majority of these forward exchanges are not forward exchanges of the type we have been discussing but rather a more sophisticated instrument known as currency swaps.

A **currency swap** is the simultaneous purchase and sale of a given amount of foreign exchange for two different value dates. Swaps are transacted between international businesses and their banks, between banks, and between governments when it is desirable to move out of one currency into another for a limited period without incurring foreign exchange risk. A common kind of swap is spot against forward. Consider a company such as Apple. Imagine Apple assembles laptop computers in the United States, but the screens are made in Japan. Apple also sells some of the finished laptops in Japan. So, like many companies, Apple both buys from and sells to Japan. Imagine Apple needs to change \$1 million into yen to pay its supplier of laptop screens today. Apple knows that in 90 days it will be paid ¥120 million by the Japanese importer that

forward exchange

When two parties agree to exchange currency and execute a deal at some specific date in the future.

forward exchange rate

The exchange rate governing a forward exchange transaction.

currency swap

Simultaneous purchase and sale of a given amount of foreign exchange for two different value dates.

MANAGEMENT FOCUS

Embraer and the Gyration of the Brazilian Real

For many years, Brazil was a country battered by persistently high inflation. As a result, the value of its currency, the real, depreciated steadily against the U.S. dollar. This changed in the early 2000s, when the Brazilian government was successful in bringing down annual inflation rates into the single digits. Lower inflation, coupled with policies that paved the way for the expansion of the Brazilian economy, resulted in a steady appreciation of the real against the U.S. dollar. In May 2004, 1 real bought \$0.3121; by August 2008, 1 real bought \$0.65, an appreciation of more than 100 percent.

The appreciation of the real against the dollar was a mixed bag for Embraer, the world's largest manufacturer of regional jets of up to 110 seats and one of Brazil's most prominent industrial companies. Embraer purchases many of the parts that go into its jets, including the engines and electronics, from U.S. manufacturers. As the real appreciated against the dollar, these parts cost less when translated into reals, which benefited Embraer's profit margins. However, the company also prices its aircraft in U.S. dollars, as do all manufacturers in the global market for commercial jet aircraft. So, as the real appreciated against the dollar, Embraer's dollar revenues were compressed when exchanged back into reals.

To try to deal with the impact of currency appreciation on its revenues, Embraer started in the mid-2000s to hedge against future appreciation of the real by buying forward contracts (forward contracts give the holder the right to exchange one currency—in this case, dollars—for another—in this case, reals—at some point in the future at a predetermined exchange rate). If the real had continued

to appreciate, this would have been a great strategy for Embraer because the company could have locked in the rate at which sales made in dollars were exchanged back into reals. Unfortunately for Embraer, as the global financial crisis unfolded in 2008, investors fled to the dollar, which they viewed as a safe haven, and the real *depreciated* against the dollar. Between August 2008 and November 2008, the value of the real fell by almost 40 percent against the dollar. But for the hedging, this depreciation would have actually increased Embraer's revenues in reals. Embraer, however, had locked itself into a much higher real/dollar exchange rate, and the company was forced to take a \$121 million loss on what was essentially a bad currency bet.

Following the shock of 2008, Embraer cut back on currency hedging, and most of its dollar sales and purchases are not hedged. This made Embraer's sales revenues very sensitive to the real/dollar exchange rate. By 2010, the Brazilian real was once more appreciating against the U.S. dollar, which pressured Embraer's revenues. By 2012, however, the Brazilian economy was stagnating, while inflation was starting to increase again. This led to a sustained fall in the value of the real, which fell from 1 real = \$0.644 in July 2011 to 1 real = \$0.32 by February 2017, a depreciation of 50 percent. What was bad for the Brazilian currency, however, was good for Embraer, whose stock price surged to the highest price since February 2008 on speculation that the decline on the real would lead to a boost in Embraer's revenues when expressed in reals.

Sources: D. Godoy, "Embraer Rallies as Brazilian Currency Weakens," *Bloomberg*, May 31, 2013; K. Kroll, "Embraer Fourth Quarter Profits Plunge 44% on Currency Woes," *Cleveland.com*, March 27, 2009; "A Fall from Grace: Brazil's Mediocre Economy," *The Economist*, June 8, 2013; "Brazil's Economy: The Deterioration," *The Economist*, December 7, 2013.

buys its finished laptops. It will want to convert these yen into dollars for use in the United States. Let us say today's spot exchange rate is $\$1 = ¥120$ and the 90-day forward exchange rate is $\$1 = ¥110$. Apple sells \$1 million to its bank in return for ¥120 million. Now Apple can pay its Japanese supplier. At the same time, Apple enters into a 90-day forward exchange deal with its bank for converting ¥120 million into dollars. Thus, in 90 days Apple will receive \$1.09 million ($¥120 \text{ million} / 110 = \1.09 million). Because the yen is trading at a premium on the 90-day forward market, Apple ends up with more dollars than it started with (although the opposite could also occur). The swap deal is just like a conventional forward deal in one important respect: It enables Apple to insure itself against foreign exchange risk. By engaging in a swap, Apple knows today that the ¥120 million payment it will receive in 90 days will yield \$1.09 million.



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The Nature of the Foreign Exchange Market

The foreign exchange market is not located in any one place. It is a global network of banks, brokers, and foreign exchange dealers connected by electronic communications systems. When companies wish to convert currencies, they typically go through their own banks rather than entering the market directly. The foreign exchange market has been growing at a rapid pace, reflecting a general growth in the volume of cross-border trade and investment (see Chapter 1). In March 1986, the average total value of global foreign exchange trading was about \$200 billion per day. By April 2022, the last date for which we have solid data, it had hit \$7.5 trillion a day.⁶ The most important trading centers are London (38 percent of activity); New York (19 percent of activity); and Tokyo, Hong Kong, and Singapore (all with around 4 to 8 percent of activity).⁷

London's dominance in the foreign exchange market is due to both history and geography. As the capital of the world's first major industrial trading nation, London had become the world's largest center for international banking by the end of the nineteenth century, a position it has retained. Today, London's central position between Tokyo and Singapore to the east, and New York to the west, has made it the critical link between the East Asian and New York markets. Due to the particular differences in time zones, London opens soon after Tokyo closes for the night and is still open for the first few hours of trading in New York. It is an open question, however, as to how the decision to exit from the EU (Brexit) will affect London's dominant position as a global trading center. The evidence so far is that London has seen some moderate trading decline but remains dominant in foreign exchange trading.⁸

Two features of the foreign exchange market are of particular note. The first is that the market never sleeps. Tokyo, London, and New York are all shut for only 3 hours out of every 24. During these three hours, trading continues in a number of minor centers, particularly San Francisco and Sydney, Australia. The second feature of the market is the integration of the various trading centers. High-speed computer linkages among trading centers around the globe have effectively created a single market. The integration of financial centers implies that no significant difference can exist in exchange rates quoted in the trading centers. For example, if the yen/dollar exchange rate quoted in London at 3 p.m. is $¥120 = \$1$, the yen/dollar exchange rate quoted in New York at the same time (10 a.m. New York time) will be identical. If the New York yen/dollar exchange rate were $¥125 = \$1$, a dealer could make a profit through **arbitrage**, buying a currency low and selling it high. For example, if the prices differed in London and New York as given, a dealer in New York could take \$1 million and use that to purchase ¥125 million. The dealer could then immediately sell the ¥125 million for dollars in London, where the transaction would yield \$1.041666 million, allowing the trader to book a profit of \$41,666 on the transaction. If all dealers tried to cash in on the opportunity, however, the demand for yen in New York would rise, resulting in an appreciation of the yen against the dollar such that the price differential between New York and London would quickly disappear. Because foreign exchange dealers are always watching their computer screens for arbitrage opportunities, the few that arise tend to be small, and they disappear in minutes.

Another feature of the foreign exchange market is the important role played by the U.S. dollar. Although a foreign exchange transaction can involve any two currencies, most transactions involve dollars on one side. This is true even when a dealer wants to sell a non-dollar currency and buy another. A dealer wishing to sell Mexican pesos for Japanese yen, for example, will usually sell the pesos for dollars and then use the dollars to buy yen. Although this may seem a roundabout way of doing things, it is actually cheaper than trying to find a holder of pesos who wants to buy yen. Because the volume of international transactions involving dollars is so great, finding dealers who wish to trade dollars for pesos or yen is not hard.

Due to its central role in so many foreign exchange deals, the dollar is a vehicle currency. In 2022, 88 percent of all foreign exchange transactions involved dollars on one side of the transaction. After the dollar, the most important vehicle currencies were the euro (31 percent), the Japanese yen (17 percent), and the British pound (13 percent)—reflecting the historical importance of these trading entities in the world economy.

arbitrage

The purchase of securities in one market for immediate resale in another to profit from a price discrepancy.



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Economic Theories of Exchange Rate Determination



Understand the different theories explaining how currency exchange rates are determined and their relative merits.

At the most basic level, exchange rates are determined by the demand and supply of one currency relative to the demand and supply of another. For example, if the demand for dollars outstrips the supply of them and if the supply of Japanese yen is greater than the demand for them, the dollar/yen exchange rate will change. The dollar will appreciate against the yen (and the yen will depreciate against the dollar). However, while differences in relative demand and supply explain the determination of exchange rates, they do so only in a superficial sense. This simple explanation does not reveal what factors underlie the demand for and supply of a currency. Nor does it tell us when the demand for dollars will exceed the supply (and vice versa) or when the supply of Japanese yen will exceed demand for them (and vice versa). Neither does it show under what conditions a currency is in demand or under what conditions it is not demanded. In this section, we will review economic theory's answers to these questions. This will give us a deeper understanding of how exchange rates are determined.

If we understand how exchange rates are determined, we may be able to forecast exchange rate movements. Because future exchange rate movements influence export opportunities, the profitability of international trade and investment deals, and the price competitiveness of foreign imports, this is valuable information for an international business. Unfortunately, no simple explanation is available. The forces that determine exchange rates are complex, and no theoretical consensus exists, even among academic economists who study the phenomenon every day. Nonetheless, most economic theories of exchange rate movements seem to agree that three factors have an important impact on future exchange rate movements in a country's currency: the country's price inflation, its interest rate, and market psychology.⁹

PRICES AND EXCHANGE RATES

To understand how prices are related to exchange rate movements, we first need to discuss an economic proposition known as the law of one price. Then we will discuss the theory of purchasing power parity (PPP), which links changes in the exchange rate between two countries' currencies to changes in the countries' price levels.

The Law of One Price

The **law of one price** states that in competitive markets free of transportation costs and barriers to trade (such as tariffs), identical products sold in different countries must sell for the

law of one price

In competitive markets free of transportation costs and barriers to trade, identical products sold in different countries must sell for the same price when their price is expressed in the same currency.

same price when their price is expressed in terms of the same currency.¹⁰ For example, if the exchange rate between the British pound and the dollar is $\text{£}1 = \$2$, a jacket that retails for \$80 in New York should sell for £40 in London (because $\$80/\$2 = \text{£}40$). Consider what would happen if the jacket cost £30 in London (\$60 in U.S. currency). At this price, it would pay a trader to buy jackets in London and sell them in New York (an example of *arbitrage*). The company initially could make a profit of \$20 on each jacket by purchasing it for £30 (\$60) in London and selling it for \$80 in New York (we are assuming away transportation costs and trade barriers). However, the increased demand for jackets in London would raise their price in London, and the increased supply of jackets in New York would lower their price there. This would continue until prices were equalized. Thus, prices might equalize when the jacket cost £35 (\$70) in London and \$70 in New York (assuming no change in the exchange rate of $\text{£}1 = \$2$).

Purchasing Power Parity

If the law of one price were true for all goods and services, the *purchasing power parity* (PPP) exchange rate could be found from any individual set of prices. By comparing the prices of identical products in different currencies, it would be possible to determine the “real,” or PPP, exchange rate that would exist if markets were efficient. (An **efficient market** has no impediments to the free flow of goods and services, such as trade barriers.)

A less extreme version of the PPP theory states that given relatively efficient markets—that is, markets in which few impediments to international trade exist—the price of a “basket of goods” should be roughly equivalent in each country. To express the PPP theory in symbols, let $P_{\$}$ be the U.S. dollar price of a basket of particular goods and $P_{¥}$ be the price of the same basket of goods in Japanese yen. The PPP theory predicts that the dollar/yen exchange rate, $E_{\$/¥}$, should be equivalent to

$$E_{\$/¥} = P_{\$} / P_{¥}$$

efficient market

A market where prices reflect all available information.

Thus, if a basket of goods costs \$200 in the United States and ¥20,000 in Japan, PPP theory predicts that the dollar/yen exchange rate should be $\$200/¥20,000$ or \$0.01 per Japanese yen (i.e., $\$1 = ¥100$).

Every year, the news magazine *The Economist* publishes its own version of the PPP theorem, which it refers to as the “Big Mac Index.” *The Economist* has selected McDonald’s Big Mac as a proxy for a “basket of goods” because it is produced according to more or less the same recipe in about 120 countries. The Big Mac PPP is the exchange rate that would have hamburgers costing the same in each country. According to *The Economist*, comparing a country’s actual exchange rate with the one predicted by the PPP theorem based on the relative prices of Big Macs is a test of whether a currency is undervalued or not. This is not a totally serious exercise, as *The Economist* admits, but it does provide a useful illustration of the PPP theorem.

To calculate the index, *The Economist* converts the price of a Big Mac in a country into dollars at current exchange rates and divides that by the average price of a Big Mac in America. According to the PPP theorem, the prices should be the same. If they are not, it implies that the currency is either overvalued against the dollar or undervalued. For example, in December 2022, the average price of a Big Mac in the United States was \$5.36, while it was 24.00 yuan in China (or \$3.54 at the December 2022 exchange rate of $\$1 = 6.78$ yuan). This suggests that the Chinese yuan is 34 percent undervalued against the U.S. dollar!

The next step in the PPP theory is to argue that the exchange rate will change if relative prices change. For example, imagine there is no price inflation in the United States, while prices in Japan are increasing by 10 percent a year. At the beginning of the year, a basket of goods costs \$200 in the United States and ¥20,000 in Japan, so the dollar/yen exchange rate, according to PPP theory, should be $\$1 = ¥100$. At the end of the year, the basket of goods still costs \$200 in the United States, but it costs ¥22,000 in Japan. PPP theory predicts that the exchange rate should change as a result. More precisely, by the end of the year

$$E_{\$/¥} = \$200/¥22,000$$

Thus, ¥1 = \$0.0091 (or \$1 = ¥110). Because of 10 percent price inflation, the Japanese yen has depreciated by 10 percent against the dollar. One dollar will buy 10 percent more yen at the end of the year than at the beginning.

Money Supply and Price Inflation

In essence, PPP theory predicts that changes in relative prices will result in a change in exchange rates. Theoretically, a country in which price inflation is running wild should expect to see its currency depreciate against that of countries where inflation rates are lower. A good example of this can be found in the opening case, where we saw that the value of the Türk lira against the U.S. dollar depreciated rapidly in 2021 due to the acceleration of price inflation in Türkiye.

If we can predict what a country's future inflation rate is likely to be, we can also predict how the value of its currency relative to other currencies—its exchange rate—is likely to change. The growth rate of a country's money supply determines its likely future inflation rate.¹¹ Thus, in theory at least, we can use information about the growth in money supply to forecast exchange rate movements.

Inflation is a monetary phenomenon. It occurs when the quantity of money in circulation rises faster than the stock of goods and services—that is, when the money supply increases faster than output increases. Imagine what would happen if everyone in the country was suddenly given \$10,000 by the government. Many people would rush out to spend their extra money on those things they had always wanted—new cars, new furniture, better clothes, and so on. Demand for goods and services would surge. Car dealers, department stores, and other providers of goods and services would respond to this upsurge in demand by raising prices. The result would be price inflation. If this example sounds familiar, that is because something similar to this happened in the United States and many other countries in 2020–2022 when national governments pumped money into their economies to prop up demand in the face of the COVID pandemic. In the case of the United States, in 2020 and 2021 the government provided a total of \$5.7 trillion in COVID-related assistance, including \$850 billion in direct payments to taxpayers.¹² Not surprisingly, in 2022 inflation surged to a 30-year high in the United States, just as the theory would predict.

A government increasing the money supply is analogous to giving people more money. An increase in the money supply makes it easier for banks to borrow from the government and for individuals and companies to borrow from banks. The resulting increase in credit causes increases in demand for goods and services. Unless the output of goods and services is growing at a rate similar to that of the money supply, the result will be inflation. This relationship has been observed time after time in country after country.

So now we have a connection between the growth in a country's money supply, price inflation, and exchange rate movements. Put simply, *when the growth in a country's money supply is faster than the growth in its output, price inflation is fueled*. The PPP theory tells us that a country with a high inflation rate will see depreciation in its currency exchange rate. You can see a current example of this in the opening case on the collapse in value of the Turkish lira. In one of the clearest historical examples, in the mid-1980s, Bolivia experienced *hyperinflation*—an explosive and seemingly uncontrollable price inflation in which money loses value very rapidly. Table 10.1 presents data on Bolivia's money supply, inflation rate, and its peso's exchange rate with the U.S. dollar during the period of hyperinflation. The exchange rate is actually the “black market” exchange rate because the Bolivian government prohibited converting the peso to other currencies during the period. The data show that the growth in money supply, the rate of price inflation, and the depreciation of the peso against the dollar all moved in step with each other. This is just what PPP theory and monetary economics predict. Between April 1984 and July 1985, Bolivia's money supply increased by 17,433 percent, prices increased by 22,908 percent, and the value of the peso against the dollar fell by 24,662 percent! In October 1985, the Bolivian government instituted a dramatic stabilization plan—which included the introduction of a new currency and tight control of the money supply—and by 1987, the country's annual inflation rate was down to 16 percent.¹³

Another way of looking at the same phenomenon is that an increase in a country's money supply, which increases the amount of currency available, changes the relative demand-and-supply conditions in the foreign exchange market. If the U.S. money supply is growing more rapidly

TABLE 10.1 Macroeconomic Data for Bolivia, April 1984 to October 1985

Month	Money Supply (billions of pesos)	Price Level Relative to 1982 (average = 1)	Exchange Rate (pesos per dollar)
1984			
April	270	21.1	3,576
May	330	31.1	3,512
June	440	32.3	3,342
July	599	34.0	3,570
August	718	39.1	7,038
September	889	53.7	13,685
October	1,194	85.5	15,205
November	1,495	112.4	18,469
December	3,296	180.9	24,515
1985			
January	4,630	305.3	73,016
February	6,455	863.3	141,101
March	9,089	1,078.6	128,137
April	12,885	1,205.7	167,428
May	21,309	1,635.7	272,375
June	27,778	2,919.1	481,756
July	47,341	4,854.6	885,476
August	74,306	8,081.0	1,182,300
September	103,272	12,647.6	1,087,440
October	132,550	12,411.8	1,120,210

Source: Juan-Antonio Morales, "Inflation Stabilization in Bolivia," in *Inflation Stabilization: The Experience of Israel, Argentina, Brazil, Bolivia, and Mexico*, ed. Michael Bruno et al. (Cambridge, MA: MIT Press, 1988).

than U.S. output, dollars will be relatively more plentiful than the currencies of countries where monetary growth is closer to output growth. As a result of this relative increase in the supply of dollars, the dollar will depreciate on the foreign exchange market against the currencies of countries with slower monetary growth.

Government policy determines whether the rate of growth in a country's money supply is greater than the rate of growth in output. A government can increase the money supply simply by telling the country's central bank to issue more money. Governments tend to do this to finance public expenditure (building roads, paying government workers, paying for defense, etc.). A government could finance public expenditure by raising taxes, but because nobody likes paying more taxes and because politicians do not like to be unpopular, they have a natural preference for expanding the money supply. Unfortunately, there is no magic money tree. The result of *excessive* growth in money supply is typically price inflation. However, this reality has not stopped governments around the world from expanding the money supply, with predictable results.

If an international business is attempting to predict future movements in the value of a country's currency on the foreign exchange market, it should examine that country's policy toward monetary growth. If the government seems committed to controlling the rate of growth



In the 1980s, the value of the Bolivian currency, the Boliviano, depreciated rapidly due to hyperinflation.
Andrzej Rostek/Shutterstock

in money supply, the country's future inflation rate may be low (even if the current rate is high) and its currency should not depreciate too much on the foreign exchange market. If the government seems to lack the political will to control the rate of growth in money supply, the future inflation rate may be high, which is likely to cause its currency to depreciate. Historically, many Latin American governments have fallen into this latter category, including Argentina, Bolivia, Brazil, and most recently, Venezuela, where uncontrolled growth in the money supply led to the inflation rate hitting 1 million percent in 2018 (i.e., the currency was essentially worthless). In late 2010, when the U.S. Federal Reserve decided to promote growth by expanding the U.S. money supply using a technique known as quantitative easing, critics charged that this too would lead to inflation and a decline in the value of the U.S. dollar on foreign exchange markets, but were they right? For a discussion of this, see the accompanying Country Focus.

Empirical Tests of PPP Theory

PPP theory predicts that exchange rates are determined by relative prices and that changes in relative prices will result in a change in exchange rates. A country in which price inflation is

COUNTRY FOCUS

Quantitative Easing, Inflation, and the Value of the U.S. Dollar

In fall 2010, the U.S. Federal Reserve (the Fed) decided to expand the U.S. money supply by entering the open

market and purchasing \$600 billion in U.S. government bonds from bondholders, a technique known as *quantitative easing*. Where did the \$600 billion come from? The Fed simply created new bank reserves and used this cash to pay for the bonds. It had, in effect, printed money. The Fed took this action in an attempt to stimulate the U.S. economy, which, in the aftermath of the 2008–2009 global financial crisis, was struggling with low economic growth

and high unemployment rates. The Fed had already tried to stimulate the economy by lowering short-term interest rates, but these were already close to zero, so it decided to lower medium- to longer-term rates; its tool for doing this was to pump \$600 billion into the economy, increasing the supply of money and lowering its price, the interest rate. The Fed pursued further rounds of quantitative easing in 2011 through to 2013. In 2014, with the U.S. economy getting stronger and unemployment falling below 6 percent, the Fed progressively reduced its bond buying program. It ended the program in October 2014. By that time, the Fed had effectively pumped more than \$3.5 trillion into the U.S. economy.

Critics were quick to attack the Fed's moves. Many claimed that the policy of expanding the money supply would fuel inflation and lead to a decline in the value of the U.S. dollar on the foreign exchange market. Some even called the policy a deliberate attempt by the Fed to debase the value of the U.S. currency, thereby driving down its value and promoting U.S. exports, which, if true, would be a form of mercantilism.

However, these charges may be unfounded for two reasons. First, at the time, the core U.S. inflation rate was the lowest in 50 years. In fact, the Fed actually feared the risk of deflation (a persistent fall in prices), which is a very damaging phenomenon. When prices are falling, people hold off their purchases because they know that goods will be cheaper tomorrow than they are today. This pattern can result in a collapse in aggregate demand and high unemployment. The Fed felt that a little inflation—say, 2 percent per year—might be a good thing. Second, U.S. economic

growth had been weak, unemployment was high, and excess productive capacity existed in the economy. Consequently, if the injection of money into the economy did stimulate demand, this would not translate into price inflation because the first response of businesses would be to expand output to utilize their excess capacity. Defenders of the Fed argued that the important point, which the critics seemed to be missing, was that expanding the money supply leads only to higher price inflation when unemployment is relatively low and the economy does not have much excess capacity, a situation that did not exist in fall 2010. As for the currency market, its reaction was muted. At the beginning of November 2010, just before the Fed announced its policy, a trade-weighted index of the value of the dollar against a basket of other major currencies stood at 72. At the end of January 2014, it stood at 78—a slight appreciation. In short, currency traders did not seem to be selling off the dollar or reflecting worries about high inflation rates.

By March 2016, with the program over, no surge in price inflation in the U.S. economy occurred. Indeed, inflation rates remained near historic lows. Moreover, far from weakening, the U.S. dollar had increased in value against most currencies, and the index value stood at 92. The Fed, it would seem, had been right and the critics were wrong.

Sources: P. Wallsten and S. Reddy, "Fed's Bond Buying Plan Ignites Growing Criticism," *The Wall Street Journal*, November 15, 2010; S. Chan, "Under Attack, the Fed Defends Policy of Buying Bonds," *International Herald Tribune*, November 17, 2010; "What QE Means for the World: Positive Sum Currency Wars," *The Economist*, February 14, 2013.

running wild should expect to see its currency depreciate against that of countries with lower inflation rates. This is intuitively appealing, but is it true in practice? Several good examples of the connection between a country's price inflation and exchange rate position (such as Bolivia) exist. However, extensive empirical testing of PPP theory has yielded mixed results.¹⁴ While PPP theory seems to yield relatively accurate predictions in the long run, it does not appear to be a strong predictor of short-run movements in exchange rates covering time spans of five years or less.¹⁵ In addition, the theory seems to best predict exchange rate changes for countries with high rates of inflation and underdeveloped capital markets. The theory is less useful for predicting short-term exchange rate movements between the currencies of advanced industrialized nations that have relatively small differentials in inflation rates.

The failure to find a strong link between relative inflation rates and exchange rate movements has been referred to as the purchasing power parity puzzle. Several factors may explain the failure of PPP theory to predict exchange rates more accurately.¹⁶ PPP theory assumes away transportation costs and barriers to trade. In practice, these factors are significant, and they tend to create significant price differentials between countries. Transportation costs are certainly not trivial for many goods. Moreover, as we saw in Chapter 7, governments routinely intervene in international trade, creating tariff and nontariff barriers to cross-border trade. Barriers to trade limit the ability of traders to use arbitrage to equalize prices for the same product in different countries, which is required for the law of one price to hold. Government intervention in cross-border trade, by violating the assumption of efficient markets, weakens the link between relative price changes and changes in exchange rates predicted by PPP theory.

PPP theory may not hold if many national markets are dominated by a handful of multinational enterprises that have sufficient market power to be able to exercise some influence over

prices, control distribution channels, and differentiate their product offerings between nations.¹⁷ In fact, this situation seems to prevail in a number of industries. In such cases, dominant enterprises may be able to exercise a degree of pricing power, setting different prices in different markets to reflect varying demand conditions. This is referred to as price discrimination. For price discrimination to work, arbitrage must be limited. According to this argument, enterprises with some market power may be able to control distribution channels and therefore limit the unauthorized resale (arbitrage) of products purchased in another national market. They may also be able to limit resale (arbitrage) by differentiating otherwise identical products among nations along some line, such as design or packaging.

For example, even though the version of Microsoft Office sold in China may be less expensive than the version sold in the United States, the use of arbitrage to equalize prices may be limited because few Americans would want a version that was based on Chinese characters. The design differentiation between Microsoft Office for China and for the United States means that the law of one price would not work for Microsoft Office, even if transportation costs were trivial and tariff barriers between the United States and China did not exist. If the inability to practice arbitrage were widespread enough, it would break the connection between changes in relative prices and exchange rates predicted by the PPP theorem and help explain the limited empirical support for this theory.

Another factor of some importance is that governments also intervene in the foreign exchange market in an attempt to influence the value of their currencies. We look at why and how they do this in Chapter 11. For now, the important thing to note is that governments regularly intervene in the foreign exchange market, and this further weakens the link between price changes and changes in exchange rates. One more factor explaining the failure of PPP theory to predict short-term movements in foreign exchange rates is the impact of investor psychology and other factors on currency purchasing decisions and exchange rate movements. We discuss this issue in more detail later in this chapter.

INTEREST RATES AND EXCHANGE RATES

Economic theory tells us that interest rates reflect expectations about likely future inflation rates. In countries where inflation is expected to be high, interest rates also will be high, because investors want compensation for the decline in the value of their money. This relationship was first formalized by economist Irvin Fisher and is referred to as the Fisher effect. The **Fisher effect** states that a country's "nominal" interest rate (i) is the sum of the required "real" rate of interest (r) and the expected rate of inflation over the period for which the funds are to be lent (π). More formally,

$$i \approx r + \pi$$

For example, if the real rate of interest in a country is 5 percent and annual inflation is expected to be 10 percent, the nominal interest rate will be around 15 percent. As predicted by the Fisher effect, a strong relationship seems to exist between inflation rates and interest rates.¹⁸

We can take this one step further and consider how it applies in a world of many countries and unrestricted capital flows. When investors are free to transfer capital between countries, real interest rates will be the same in every country. If differences in real interest rates did emerge between countries, arbitrage would soon equalize them. For example, if the real interest rate in Japan was 10 percent and only 6 percent in the United States, investors would profit from borrowing money in the United States and investing it in Japan. The resulting increase in the demand for money in the United States would raise the real interest rate there, while the increase in the supply of foreign money in Japan would lower the real interest rate there. This would continue until the two sets of real interest rates were equalized.

It follows from the Fisher effect that if the real interest rate is the same worldwide, any difference in interest rates between countries reflects differing expectations about inflation rates. Thus, if the expected rate of inflation in the United States is greater than that in Japan, U.S. nominal interest rates will be greater than Japanese nominal interest rates.

Because we know from PPP theory that a link exists (in theory, at least) between inflation and exchange rates and because interest rates reflect expectations about inflation, it follows that there

Fisher effect

Nominal interest rates (i) in each country equal the required real rate of interest (r) and the expected rate of inflation over the period of time for which the funds are to be lent (π). That is, $i = r + \pi$.

must also be a link between interest rates and exchange rates. This link is known as the international Fisher effect. The **international Fisher effect (IFE)** states that for any two countries, the spot exchange rate should change in an equal amount but in the opposite direction to the difference in nominal interest rates between the two countries. Stated more formally, the change in the spot exchange rate between the United States and Japan, for example, can be modeled as follows:

$$\frac{S_1 - S_2}{S_2} \times 100 = i_s - i_y$$

where i_s and i_y are the respective nominal interest rates in the United States and Japan, S_1 is the spot exchange rate at the beginning of the period, and S_2 is the spot exchange rate at the end of the period. If the U.S. nominal interest rate is higher than Japan's, reflecting greater expected inflation rates, the value of the dollar against the yen should fall by that interest rate differential in the future. So if the interest rate in the United States is 10 percent and in Japan it is 6 percent, we would expect the value of the dollar to depreciate by 4 percent against the Japanese yen.

Do interest rate differentials help predict future currency movements? The evidence is mixed; as in the case of PPP theory, in the long run, a relationship seems to exist between interest rate differentials and subsequent changes in spot exchange rates. However, considerable short-run deviations occur. Like PPP, the international Fisher effect is not a good predictor of short-run changes in spot exchange rates.¹⁹

INVESTOR PSYCHOLOGY AND BANDWAGON EFFECTS

Empirical evidence suggests that neither PPP theory nor the international Fisher effect is particularly good at explaining short-term movements in exchange rates. One reason may be the impact of investor psychology on short-run exchange rate movements. Evidence reveals that various psychological factors play an important role in determining the expectations of market traders as to likely future exchange rates.²⁰ In turn, expectations have a tendency to become self-fulfilling prophecies.

A particularly famous example of this mechanism occurred in September 1992, when the international financier George Soros made a huge bet against the British pound. Soros borrowed billions of pounds, using the assets of his investment funds as collateral, and immediately sold those pounds for German deutsche marks (this was before the advent of the euro). This technique, known as short selling, can earn the speculator enormous profits if he can subsequently buy back the pounds he sold at a much better exchange rate and then use those pounds, purchased cheaply, to repay his loan. By selling pounds and buying deutsche marks, Soros helped start pushing down the value of the pound on the foreign exchange markets. More importantly, when Soros started shorting the British pound, many foreign exchange traders, knowing Soros's reputation, jumped on the bandwagon and did likewise. This triggered a classic **bandwagon effect** with traders moving as a herd in the same direction at the same time. As the bandwagon effect gained momentum, with more traders selling British pounds and purchasing deutsche marks in expectation of a decline in the pound, their expectations became a self-fulfilling prophecy. Massive selling forced down the value of the pound against the deutsche mark. Soros earned more than £1 billion in profits from his short position, and Britain was forced to pull the pound out of the existing exchange rate agreement because its exchange rate fell below what was required to participate. In other words, the pound declined in value not so much because of any major shift in macroeconomic fundamentals but because investors followed a bet placed by a major speculator, George Soros.

According to a number of studies, investor psychology and bandwagon effects play an important role in determining short-run exchange rate movements.²¹ However, these effects can be hard to predict. Investor psychology can be influenced by political factors and by microeconomic events, such as the investment decisions of individual firms, many of which are only loosely linked to macroeconomic fundamentals, such as relative inflation rates. Also, bandwagon effects can be both triggered and exacerbated by the idiosyncratic behavior of politicians. Something like this seems to have occurred in Southeast Asia during 1997 when, one after another, the currencies of Thailand, Malaysia, South Korea, and Indonesia lost between 50 and 70 percent of their value against the U.S. dollar in a few months.

international Fisher effect (IFE)

For any two countries, the spot exchange rate should change in an equal amount but in the opposite direction to the difference in nominal interest rates between countries.

bandwagon effect

Movement of traders like a herd, all in the same direction and at the same time, in response to each other's perceived actions.

SUMMARY OF EXCHANGE RATE THEORIES

Relative monetary growth, relative inflation rates, and nominal interest rate differentials are all moderately good predictors of long-run changes in exchange rates. They are poor predictors of short-run changes in exchange rates, however, perhaps because of the impact of psychological factors, investor expectations, and bandwagon effects on short-term currency movements. This information is useful for an international business. Insofar as the long-term profitability of foreign investments, export opportunities, and the price competitiveness of foreign imports are all influenced by long-term movements in exchange rates, international businesses would be advised to pay attention to countries' differing monetary growth, inflation, and interest rates. International businesses that engage in foreign exchange transactions on a day-to-day basis could benefit by knowing some predictors of short-term foreign exchange rate movements. Unfortunately, short-term exchange rate movements are difficult to predict.



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Exchange Rate Forecasting



Identify the merits of different approaches toward exchange rate forecasting.

A company's need to predict future exchange rate variations raises the issue of whether it is worthwhile for the company to invest in exchange rate forecasting services to aid decision making. Two schools of thought address this issue. The efficient market school argues that forward exchange rates do the best possible job of forecasting future spot exchange rates and, therefore, investing in forecasting services would be a waste of money. The other school of thought, the inefficient market school, argues that companies can improve the foreign exchange market's estimate of future exchange rates (as contained in the forward rate) by investing in forecasting services. In other words, this school of thought does not believe the forward exchange rates are the best possible predictors of future spot exchange rates.

THE EFFICIENT MARKET SCHOOL

Forward exchange rates represent market participants' collective predictions of likely spot exchange rates at specified future dates. If forward exchange rates are the best possible predictor of future spot rates, it would make no sense for companies to spend additional money trying to forecast short-run exchange rate movements. Many economists believe the foreign exchange market is efficient at setting forward rates.²² An efficient market is one in which prices reflect all available public information. (If forward rates reflect all available information about likely future changes in exchange rates, a company cannot beat the market by investing in forecasting services.)



Did You Know?

Did you know that the value of the U.S. dollar fluctuates significantly against the value of the euro?

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If the foreign exchange market is efficient, forward exchange rates should be unbiased predictors of future spot rates. This does not mean the predictions will be accurate in any specific situation. It means inaccuracies will not be consistently above or below future spot rates; they will be random. Many empirical tests have addressed the efficient market hypothesis. Although most of the early work seems to confirm the hypothesis (suggesting that companies should not waste their money on forecasting services), some studies have challenged it.²³ Some evidence exists that forward rates are not unbiased predictors of future spot rates and that more accurate predictions of future spot rates can be calculated from publicly available information.²⁴

THE INEFFICIENT MARKET SCHOOL

Citing evidence against the efficient market hypothesis, some economists believe the foreign exchange market is inefficient. An **inefficient market** is one in which prices do not reflect all available information. In an inefficient market, forward exchange rates will not be the best possible predictors of future spot exchange rates.

If this is true, it may be worthwhile for international businesses to invest in forecasting services (as many do). The belief is that professional exchange rate forecasts might provide better predictions of future spot rates than forward exchange rates do. However, the track record of professional forecasting services is not very good.²⁵ For example, forecasting services did not predict the 1997 currency crisis that swept through Southeast Asia, nor did they predict the rise in the value of the dollar that occurred during late 2008, a period when the United States fell into a deep financial crisis that some thought would lead to a decline in the value of the dollar (it appears that the dollar rose because it was seen as a relatively safe currency in a time when many nations were experiencing economic trouble).

inefficient market

One in which prices do not reflect all available information.

APPROACHES TO FORECASTING

Assuming the inefficient market school is correct that the foreign exchange market's estimate of future spot rates can be improved, on what basis should forecasts be prepared? Here again, two schools of thought exist. One adheres to fundamental analysis, while the other uses technical analysis.

Fundamental Analysis

Fundamental analysis draws on economic theory to construct sophisticated econometric models for predicting exchange rate movements. The variables contained in these models typically include those we have discussed, such as relative money supply growth rates, inflation rates, and interest rates. In addition, they may include variables related to balance-of-payments positions.

Running a deficit on a balance-of-payments current account (a country is importing more goods and services than it is exporting) creates pressures that may result in the depreciation of the country's currency on the foreign exchange market.²⁶ Consider what might happen if the United States were running a persistent current account balance-of-payments deficit (as it has been). Because the United States would be importing more than it was exporting, people in other countries would be increasing their holdings of U.S. dollars. If these people were willing to hold their dollars, the dollar's exchange rate would not be influenced. However, if these people

converted their dollars into other currencies, the supply of dollars in the foreign exchange market would increase (as would demand for the other currencies). This shift in demand and supply would create pressures that could lead to the depreciation of the dollar against other currencies.

This argument hinges on whether people in other countries are willing to hold dollars. This willingness depends on such factors as U.S. interest rates, the return on holding other dollar-denominated assets such as stocks in U.S. companies, and, most importantly, inflation rates. So, in a sense, the balance-of-payments situation is not a fundamental predictor of future exchange rate movements. But what makes financial assets such as stocks and bonds attractive? The answer is prevailing interest rates and inflation rates, both of which affect underlying economic growth and the real return to holding U.S. financial assets. Given this, we are back to the argument that the fundamental determinants of exchange rates are monetary growth, inflation rates, and interest rates.

Technical Analysis

Technical analysis uses price and volume data to determine past trends, which are expected to continue into the future. This approach does not rely on a consideration of economic fundamentals. Technical analysis is based on the premise that analyzable market trends and waves exist and that previous trends and waves can be used to predict future trends and waves. Because no theoretical rationale has been found for this assumption of predictability, many economists compare technical analysis to fortune-telling. Despite this skepticism, technical analysis continues to be popular.²⁷



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freely convertible currency

A country's currency is freely convertible when the government of that country allows both residents and nonresidents to purchase unlimited amounts of foreign currency with the domestic currency.

externally convertible currency

Limitations on the ability of residents to convert domestic currency, though nonresidents can convert their holdings of domestic currency into foreign currency.

nonconvertible currency

A currency is not convertible when both residents and nonresidents are prohibited from converting their holdings of that currency into another currency.

Currency Convertibility

Until this point, we have assumed that the currencies of various countries are freely convertible into other currencies. Due to government restrictions, a significant number of currencies are not freely convertible into other currencies. A country's currency is said to be **freely convertible** when the country's government allows both residents and nonresidents to purchase unlimited amounts of a foreign currency with it. A currency is said to be **externally convertible** when only nonresidents may convert it into a foreign currency without any limitations. A currency is **nonconvertible** when neither residents nor nonresidents are allowed to convert it into a foreign currency.

Free convertibility is not universal. Many countries place some restrictions on their residents' ability to convert the domestic currency into a foreign currency (a policy of external convertibility). Restrictions range from the relatively minor (such as restricting the amount of foreign currency residents may take with them out of the country on trips) to the major (such as restricting domestic businesses' ability to take foreign currency out of the country). External convertibility restrictions can limit domestic companies' ability to invest abroad, but they present few problems for foreign companies wishing to do business in that country. For example, even if the Japanese government tightly controlled the ability of its residents to convert the yen into U.S. dollars, all U.S. businesses with deposits in Japanese banks may at any time convert all their yen into dollars and take them out of the country. Thus, a U.S. company with a subsidiary in Japan is assured that it will be able to convert the profits from its Japanese operation into dollars and take them out of the country.

Serious problems arise, however, under a policy of nonconvertibility. This was the practice of the former Soviet Union, and it continued to be the practice in Russia for several years after the collapse of the Soviet Union. When strictly applied, nonconvertibility means that although a U.S. company doing business in a country such as Russia may be able to generate significant

ruble profits, it may not convert those rubles into dollars and take them out of the country. Obviously, this is not desirable for international business.

Governments limit convertibility to preserve their foreign exchange reserves. A country needs an adequate supply of these reserves to service its international debt commitments and to purchase imports. Governments typically impose convertibility restrictions on their currency when they fear that free convertibility will lead to a run on their foreign exchange reserves. This occurs when residents and nonresidents rush to convert their holdings of domestic currency into a foreign currency—a phenomenon generally referred to as **capital flight**. Capital flight is most likely to occur when the value of the domestic currency is depreciating rapidly because of hyperinflation or when a country's economic prospects are shaky in other respects. Under such circumstances, both residents and nonresidents tend to believe that their money is more likely to hold its value if it is converted into a foreign currency and invested abroad. Not only will a run on foreign exchange reserves limit the country's ability to service its international debt and pay for imports, but it will also lead to a precipitous depreciation in the exchange rate as residents and nonresidents unload their holdings of domestic currency on the foreign exchange markets (thereby increasing the market supply of the country's currency). Governments fear that the rise in import prices resulting from currency depreciation will lead to further increases in inflation. This fear provides another rationale for limiting convertibility.

capital flight

Converting domestic currency into a foreign currency.

Companies can deal with the nonconvertibility problem by engaging in countertrade. **Countertrade** refers to a range of barter-like agreements by which goods and services can be traded for other goods and services. Countertrade can make sense when a country's currency is nonconvertible. For example, consider the deal that General Electric struck with the Romanian government when that country's currency was nonconvertible. When General Electric won a contract for a \$150 million generator project in Romania, it agreed to take payment in the form of Romanian goods that could be sold for \$150 million on international markets. In a similar case, the Venezuelan government negotiated a contract with Caterpillar under which Venezuela would trade 350,000 tons of iron ore for Caterpillar heavy construction equipment. Caterpillar subsequently traded the iron ore to Romania in exchange for Romanian farm products, which it then sold on international markets for dollars.²⁸

countertrade

The trade of goods and services for other goods and services.

How important is countertrade? Twenty years ago, a large number of nonconvertible currencies existed in the world, and countertrade was quite significant. However, in recent years, many governments have made their currencies freely convertible, and the percentage of world trade that involves countertrade is probably significantly below 5 percent.²⁹

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**360° View: Managerial Implications****LO10-6**

Compare and contrast the differences among translation, transaction, and economic exposure, and explain the implications for management practice.

FOREIGN EXCHANGE RATE RISK

This chapter contains a number of clear implications for business. First, it is critical that international businesses understand the influence of exchange rates on the profitability of trade and investment deals. Adverse changes in exchange rates can make apparently profitable deals unprofitable. As noted, the risk introduced into international business transactions by changes in exchange rates is referred to as foreign exchange risk. Foreign exchange risk is usually divided into three main categories: transaction exposure, translation exposure, and economic exposure.

Transaction Exposure

transaction exposure

The extent to which income from individual transactions is affected by fluctuations in foreign exchange values.

Transaction exposure is the extent to which the income from individual transactions is affected by fluctuations in foreign exchange values. Such exposure includes obligations for the purchase or sale of goods and services at previously agreed prices and the borrowing or lending of funds in foreign currencies. For example, suppose in 2014, an American airline agreed to purchase 10 Airbus 330 aircraft for €120 million each for a total price of €1.20 billion, with delivery scheduled for 2018 and payment due then. When the contract was signed in 2014, the dollar/euro exchange rate stood at $\$1 = €1.10$, so the American airline anticipated paying \$1.09 billion for the 10 aircraft when they were delivered ($€1.2 \text{ billion} / 1.1 = \1.09 billion). However, imagine that the value of the dollar depreciates against the euro over the intervening period, so that a dollar buys only €0.80 in 2018 when payment was due ($\$1 = €0.80$). Now the total cost in U.S. dollars is \$1.5 billion ($€1.2 \text{ billion} / 0.80 = \1.5 billion), an increase of \$0.41 billion! The transaction exposure here is \$0.41 billion, which is the money lost due to an adverse movement in exchange rates between the time when the deal was signed and when the aircraft were paid for.

Translation Exposure

translation exposure

The extent to which the reported consolidated results and balance sheets of a corporation are affected by fluctuations in foreign exchange values.

Translation exposure is the impact of currency exchange rate changes on the reported financial statements of a company. Translation exposure is concerned with the present measurement of past events. The resulting accounting gains or losses are said to be unrealized—they are “paper” gains and losses—but they are still important. Consider a U.S. firm with a subsidiary in Mexico. If the value of the Mexican peso depreciates significantly against the dollar, this would substantially reduce the dollar value of the Mexican subsidiary’s equity. In turn, this would reduce the total dollar value of the firm’s equity reported in its consolidated balance sheet. This would raise the apparent leverage of the firm (its debt ratio), which could increase the firm’s cost of borrowing and potentially limit its access to the capital market. Similarly, if an American firm has a subsidiary in the European Union and the value of the euro depreciates rapidly against that of the dollar over a year, this will reduce the dollar value of the euro profit made by the European subsidiary, resulting in negative translation exposure. In fact, many U.S. firms suffered from significant negative translation exposure in Europe during 2000, precisely because the euro did depreciate rapidly against the dollar. In 2002–2007, the euro rose in value against the dollar. This positive translation exposure boosted the dollar profits of American multinationals with significant operations in Europe. Between mid-2014 and early 2015, the euro slumped in value against the dollar, compressing the dollar profits of American multinationals with significant European exposure.

Economic Exposure

economic exposure

The extent to which a firm’s future international earning power is affected by changes in exchange rates.

Economic exposure is the extent to which a firm’s future international earning power is affected by changes in exchange rates. Economic exposure is concerned with the long-run effect of changes in exchange rates on future prices, sales, and costs. This is distinct from transaction exposure, which is concerned with the effect of exchange rate changes on individual transactions, most of which are short-term affairs that will be executed within a few weeks or months. Consider the effect of wide swings in the value of the dollar on many U.S. firms’ international competitiveness. The rapid rise in the value of the dollar on the foreign exchange market in the 1990s hurt the price competitiveness of many U.S. producers in world markets. U.S. manufacturers that relied heavily on exports saw their export volume and world market share decline. The reverse phenomenon occurred in 2000–2009, when the dollar declined against most major currencies.

The fall in the value of the dollar helped increase the price competitiveness of U.S. manufacturers in world markets. Between mid-2014 and early 2015, the dollar increased significantly in value against most major currencies, decreasing the price competitiveness of U.S. exporters. In 2017, the opposite occurred, with the dollar falling in value, increasing the price competitiveness of U.S. exporters, but in 2018 to 2022 the dollar reversed course again, increasing in value and making U.S. exports more expensive.

REDUCING TRANSLATION AND TRANSACTION EXPOSURE

A number of tactics can help firms minimize their transaction and translation exposure. These tactics primarily protect short-term cash flows from adverse changes in exchange rates. We have already discussed two of these tactics at length in the chapter, entering into forward exchange rate contracts and buying swaps. In addition to buying forward and using swaps, firms can minimize their foreign exchange exposure through leading and lagging payables and receivables—that is, paying suppliers and collecting payment from customers early or late depending on expected exchange rate movements. A **lead strategy** involves attempting to collect foreign currency receivables (payments from customers) early when a foreign currency is expected to depreciate and paying foreign currency payables (to suppliers) before they are due when a currency is expected to appreciate. A **lag strategy** involves delaying collection of foreign currency receivables if that currency is expected to appreciate and delaying payables if the currency is expected to depreciate. Leading and lagging involve accelerating payments from weak-currency to strong-currency countries and delaying inflows from strong-currency to weak-currency countries.

Lead and lag strategies can be difficult to implement, however. The firm must be in a position to exercise some control over payment terms. Firms do not always have this kind of bargaining power, particularly when they are dealing with important customers who are in a position to dictate payment terms. Also, because lead and lag strategies can put pressure on a weak currency, many governments limit leads and lags. For example, some countries set 180 days as a limit for receiving payments for exports or making payments for imports.

lead strategy

Collecting foreign currency receivables early when a foreign currency is expected to depreciate and paying foreign currency payables before they are due when a currency is expected to appreciate.

lag strategy

Delaying the collection of foreign currency receivables if that currency is expected to appreciate and delaying payables if that currency is expected to depreciate.

REDUCING ECONOMIC EXPOSURE

Reducing economic exposure requires strategic choices that go beyond the realm of financial management. The key to reducing economic exposure is to distribute the firm's productive assets to various locations so the firm's long-term financial well-being is not severely affected by adverse changes in exchange rates. This is a strategy that firms both large and small sometimes pursue. For example, during the 2000s, fearing that the euro would continue to strengthen against the U.S. dollar, some European firms that did significant business in the United States set up local production facilities in that market to ensure that a rising euro did not put them at a competitive disadvantage relative to their local rivals. Similarly, Toyota has production plants distributed around the world in part to make sure that a rising yen does not price Toyota cars out of local markets. Caterpillar has also pursued this strategy, setting up factories around the world that can act as a hedge against the possibility that a strong dollar will price Caterpillar's exports out of foreign markets. In 2008, 2009, 2014–2015, and 2019–2022, all periods of dollar strength, this real hedge proved to be very useful.

OTHER STEPS FOR MANAGING FOREIGN EXCHANGE RISK

A firm needs to develop a mechanism for ensuring it maintains an appropriate mix of tactics and strategies for minimizing its foreign exchange exposure. Although businesspeople don't

universally agree on the components of this mechanism, a number of common themes stand out.³⁰ First, central control of exposure is needed to protect resources efficiently and ensure that each subunit adopts the correct mix of tactics and strategies. Many companies have set up in-house foreign exchange centers. Although such centers may not be able to execute all foreign exchange deals—particularly in large, complex multinationals where myriad transactions may be pursued simultaneously—they should at least set guidelines for the firm’s subsidiaries to follow.

Second, firms should distinguish between, on one hand, transaction and translation exposure and, on the other, economic exposure. Many companies seem to focus on reducing their transaction and translation exposure and pay scant attention to economic exposure, which may have more profound long-term implications.³¹ Firms need to develop strategies for dealing with economic exposure. For example, Stanley Black & Decker, the maker of power tools, has a strategy for actively managing its economic risk. The key to Stanley Black & Decker’s strategy is flexible sourcing. In response to foreign exchange movements, Stanley Black & Decker can move production from one location to another to offer the most competitive pricing. Stanley Black & Decker manufactures in more than a dozen locations around the world—in Europe, Australia, Brazil, Mexico, and Japan. More than 50 percent of the company’s productive assets are based outside North America. Although each of Stanley Black & Decker’s factories focuses on one or two products to achieve economies of scale, considerable overlap exists. On average, the company runs its factories at no more than 80 percent capacity, so most are able to switch rapidly from producing one product to producing another or to add a product. This allows a factory’s production to be changed in response to foreign exchange movements. For example, if the dollar depreciates against other currencies, the amount of imports into the United States from overseas subsidiaries can be reduced and the amount of exports from U.S. subsidiaries to other locations can be increased.³²

Third, the need to forecast future exchange rate movements cannot be overstated, although as we saw earlier in the chapter, this is a tricky business. No model comes close to perfectly predicting future movements in foreign exchange rates. The best that can be said is that in the short run, forward exchange rates provide the best predictors of exchange rate movements, and in the long run, fundamental economic factors—particularly relative inflation rates—should be watched because they influence exchange rate movements. Some firms attempt to forecast exchange rate movements in-house; others rely on outside forecasters. However, all such forecasts are imperfect attempts to predict the future.

Fourth, firms need to establish good reporting systems so the central finance function (or in-house foreign exchange center) can regularly monitor the firm’s exposure positions. Such reporting systems should enable the firm to identify any exposed accounts, the exposed position by currency of each account, and the time periods covered.

Finally, on the basis of the information it receives from exchange rate forecasts and its own regular reporting systems, the firm should produce monthly foreign exchange exposure reports. These reports should identify how cash flows and balance sheet elements might be affected by forecasted changes in exchange rates. Management can then use the reports as a basis for adopting tactics and strategies to hedge against undue foreign exchange risks.

Surprisingly, some of the largest and most sophisticated firms don’t take such precautionary steps, exposing themselves to very large foreign exchange risks.

Key Terms

foreign exchange market, p. 304
 exchange rate, p. 305
 foreign exchange risk, p. 306
 currency speculation, p. 307
 carry trade, p. 307
 spot exchange rate, p. 308

forward exchange, p. 309
 forward exchange rate, p. 309
 currency swap, p. 309
 arbitrage, p. 311
 law of one price, p. 312
 efficient market, p. 313
 Fisher effect, p. 318
 international Fisher effect (IFE), p. 319

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 freely convertible currency, p. 322
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countertrade, p. 323
 transaction exposure, p. 324
 translation exposure, p. 324
 economic exposure, p. 324
 lead strategy, p. 325
 lag strategy, p. 325

Summary

This chapter explained how the foreign exchange market works, examined the forces that determine exchange rates, and then discussed the implications of these factors for international business. Given that changes in exchange rates can dramatically alter the profitability of foreign trade and investment deals, this is an area of major interest to international business. The chapter made the following points:

1. One function of the foreign exchange market is to convert the currency of one country into the currency of another. A second function of the foreign exchange market is to provide insurance against foreign exchange risk.
2. The spot exchange rate is the exchange rate at which a dealer converts one currency into another currency on a particular day.
3. Foreign exchange risk can be reduced by using forward exchange rates. A forward exchange rate is an exchange rate governing future transactions. Foreign exchange risk can also be reduced by engaging in currency swaps. A swap is the simultaneous purchase and sale of a given amount of foreign exchange for two different value dates.
4. The law of one price holds that in competitive markets that are free of transportation costs and barriers to trade, identical products sold in different countries must sell for the same price when their price is expressed in the same currency.
5. Purchasing power parity (PPP) theory states the price of a basket of particular goods should be roughly equivalent in each country. PPP theory predicts that the exchange rate will change if relative prices change.
6. The rate of change in countries' relative prices depends on their relative inflation rates. A country's inflation rate seems to be a function of the growth in its money supply.
7. The PPP theory of exchange rate changes yields relatively accurate predictions of long-term trends in exchange rates but not of short-term movements.
8. Interest rates reflect expectations about inflation. In countries where inflation is expected to be high, interest rates also will be high.
9. The international Fisher effect states that for any two countries, the spot exchange rate should change in an equal amount but in the opposite direction to the difference in nominal interest rates.
10. The most common approach to exchange rate forecasting is fundamental analysis. This relies on variables such as money supply growth, inflation rates, nominal interest rates, and balance-of-payments positions to predict future changes in exchange rates.
11. In many countries, the ability of residents and nonresidents to convert local currency into a foreign currency is restricted by government policy. A government restricts the convertibility of its currency to protect the country's foreign exchange reserves and to halt any capital flight.
12. Nonconvertibility of a currency makes it very difficult to engage in international trade and investment in the country. One way of coping with the nonconvertibility problem is to engage in countertrade—to trade goods and services for other goods and services.
13. The three types of exposure to foreign exchange risk are transaction exposure, translation exposure, and economic exposure.
14. Tactics that insure against transaction and translation exposure include buying forward, using currency swaps, and leading and lagging payables and receivables.
15. Reducing a firm's economic exposure requires strategic choices about how the firm's productive assets are distributed around the globe.

Critical Thinking and Discussion Questions

1. The interest rate on South Korean government securities with one-year maturity is 4 percent, and the expected inflation rate for the coming year is 2 percent. The interest rate on U.S. government securities with one-year maturity is 7 percent, and the expected rate of inflation is 5 percent. The current spot exchange rate for Korean won is $\$1 = \text{W}1,200$. Forecast the spot exchange rate one year from today. Explain the logic of your answer.
2. Two countries, Great Britain and the United States, produce just one good: beef. Suppose the price of beef in the United States is \$2.80 per pound, and in Britain it is £3.70 per pound.
 - a. According to PPP theory, what should the dollar/pound spot exchange rate be?
 - b. Suppose the price of beef is expected to rise to \$3.10 in the United States and to £4.65 in Britain. What should the one-year forward dollar/pound exchange rate be?
 - c. Given your answers to parts *a* and *b*, and given that the current interest rate in the United States is 10 percent, what would you expect the current interest rate to be in Britain?
3. Reread the Management Focus “Embraer and the gyrations of the Brazilian Real,” and then answer the following questions:
 - a. What does the recent economic history of Brazil tell you about the relationship between price inflation and exchange rates? What other factors might determine exchange rates for the Brazilian real?
 - b. Is a decline in value of the real against the U.S. dollar good for Embraer, bad for Embraer, or a mixed bag? Explain your answer.
 - c. What kind of foreign exchange rate risks is Embraer exposed to? Can Embraer reduce these risks? How?
 - d. Do you think Embraer’s decision to try to hedge against further appreciation of the real in the early 2000s was a good decision? What was the alternative?
 - e. Since 2008, Embraer has significantly reduced its dollar hedging operations. Is this wise?
 - f. Between mid-2014 and early 2015, the real depreciated significantly against the U.S. dollar. What do you think the impact was on Embraer?
4. You manufacture wine goblets. In mid-June, you receive an order for 10,000 goblets from Japan. Payment of ¥400,000 is due in mid-December. You expect the yen to rise from its present rate of $\$1 = \text{¥}130$ to $\$1 = \text{¥}100$ by December. You can borrow yen at 6 percent a year. What should you do?
5. You are the CFO of a U.S. firm whose wholly owned subsidiary in Mexico manufactures component parts for your U.S. assembly operations. The subsidiary has been financed by bank borrowings in the United States. One of your analysts told you that the Mexican peso is expected to depreciate by 30 percent against the dollar on the foreign exchange markets over the next year. What actions, if any, should you take?

globalEDGE Research Task

Use the globalEDGE™ website (globaledge.msu.edu) to complete the following exercises:

1. One of your company’s essential suppliers is located in Japan. Your company needs to make a 1-million Japanese yen payment in six months. Considering that your company primarily operates in U.S. dollars, you are assigned the task of deciding on a strategy to minimize your transaction exposure. Identify the *spot* and *forward exchange rates* between the two currencies. What factors influence your decision to use each? Which one would you choose? How many dollars must you spend to acquire the amount of yen required?

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2. Sometimes analysts use the price of specific products in different locations to compare currency valuation and purchasing power. For example, *The Economist’s* Big Mac Index compares the purchasing power parity of many countries based on the price of a Big Mac. Using Google, locate the latest edition of this index that is accessible. Identify the five countries (and their currencies) with the lowest purchasing power parity according to this classification. Which currencies, if any, are overvalued?

CLOSING CASE



Fedor Selivanov/Shutterstock

Hedging the Thai Baht

For decades now, the exchange rate between the Thai baht (THB) and the U.S. dollar has been quite volatile. Back in 2005, 40 Thai baht were needed to buy one U.S. dollar. Due to economic growth in Thailand and strong export earnings, two years later the baht had surged to THB 30 to the U.S. dollar. Since then, the exchange rate has oscillated between $\text{THB } 30 = \$1$ and $\text{THB } 36 = \$1$. In late 2020, with Thailand running a persistent surplus in its balance of payments, the baht had appreciated against the dollar, with the exchange rate reaching THB 29.81 per U.S. dollar, but by early 2023 it was back to $\text{THB } 36 = \$1$. The baht has also exhibited volatility against other regional currencies, including the Taiwan dollar, Korean won, and Indonesian rupiah.

Recognizing that a volatile exchange rate can cause problems for Thailand's exporters and importers, the Bank of Thailand has in recent years urged firms that are engaged in international trade to hedge against volatility in the value of the baht, noting that the country faces risks from trade protectionism, geopolitical events, and other uncertainties (such as the COVID-19 pandemic). Major Thai exports include computer parts, semiconductor chips, automobiles, vehicle parts, refined petroleum, rice, rubber, and gold. Strategies recommended by the bank include buying forward to lock in the value of the baht against the U.S. dollar, or other currencies, so that export earnings are less impacted by sudden unanticipated shifts in exchange rates.

For example, imagine a Thai exporter enters into a contract to sell THB 3 billion of semiconductor chips to a U.S. firm when the exchange rate is $\text{THB } 30 = \$1$. It might expect to earn \$100 million from this deal ($\text{THB } 3 \text{ billion} \times \$1/\text{THB } 30 = \$100 \text{ million}$). However, if the value of the baht then declines in value to $\text{THB } 36 = \$1$ between when the contract was signed and when the chips are delivered, now each dollar buys more baht, and the Thai exporter will only earn \$83.3 million ($\text{THB } 3 \text{ billion} \times \$1/\text{THB } 36 = \$83.3 \text{ million}$), a reduction in revenue of \$16.7 million. This shift in exchange rates would be particularly problematic if the Thai firm is paying for some of its inputs in U.S. dollars, such as the silicone to make the chips, and was counting on the U.S. dollar revenues to do that. Now it will cost more baht to buy its inputs than expected, while the company receives less in dollars than it expected for its exports.

A 2017 survey by the Bank of Thailand revealed that 72 percent of small exporters in the country had done nothing to protect themselves from exchange rate volatility. Some 59 percent of medium-sized exporters and 50 percent of large exporters also did not hedge. This meant that the value of their export earnings and the cost of their imports were exposed to unanticipated shifts in exchange rates. On average, 66 percent of all Thai exporters did not make any hedging arrangements, and only 4 percent of them had hedged fully against foreign exchange risk. These data suggest that Thai firms are taking on a high degree of foreign exchange risk.

The last few years have illustrated how exchange rates can often defy market expectations and are difficult to predict, which increases the value of a hedging strategy. When the Trump administration took office in the United States in 2017, the assumption was that the baht would weaken in value against the United States dollar. In fact, the opposite happened. The baht strengthened. The reasons included the surprising resilience of Thai exports in the face of the global economic slowdown triggered by

the COVID-19 pandemic. One explanation for this resilience is that the trade war between the United States and China started by the Trump administration resulted in U.S. enterprises switching demand for computer parts, semiconductor chips, and auto parts from Chinese producers to Thai producers. On the face of it, this switch was good for unhedged Thai exporters (their dollar earnings were greater than anticipated).

In late 2020, the U.S. Treasury Department placed Thailand on a watch list as a potential currency manipulator, raising the possibility that Thailand might be hit with U.S. trade sanctions. The argument made by the United States seemed to be that the baht should have appreciated even more against the dollar given the country's strong exports and balance of payments surplus and that Thailand was holding down its value to boost exports. The Bank of Thailand responded by stating that they only stepped into currency markets to curb the excessive volatility of the baht and that they would never use the exchange rate to gain an unfair trade advantage. Nevertheless, due to these geopolitical risks, the Bank of Thailand again emphasized the importance of currency hedging, noting that the possible imposition of trade sanctions might have an adverse impact upon the U.S. dollar/THB exchange rate.

Case Discussion Questions

1. What are the risks for Thai firms of not hedging the value of the Thai baht against other major trading currencies, such as the U.S. dollar?
2. Despite the risks associated with not hedging, why do you think so many Thai exporters chose not to hedge?
3. If you were a consultant advising small and mid-sized Thai firms on hedging strategy, what would you suggest they do, and why?
4. Do you think it's possible for Thailand to manipulate the value of its currency against the U.S. dollar, as the U.S. Treasury Department suggested in 2020? What are the challenges they would face in doing this?

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Part 4 The Global Monetary System

11

THE INTERNATIONAL MONETARY SYSTEM

Learning Objectives

After reading this chapter, you will be able to:

- LO11-1 Describe the historical development of the modern global monetary system.
- LO11-2 Explain the role played by the World Bank and the IMF in the international monetary system.
- LO11-3 Compare and contrast the differences between a fixed and a floating exchange rate system.
- LO11-4 Identify exchange rate regimes used in the world today and why countries adopt different exchange rate regimes.
- LO11-5 Understand the debate surrounding the role of the IMF in the management of financial crises.
- LO11-6 Explain the implications of the global monetary system for management practice.



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opening case

PAKISTAN AND THE IMF

Pakistan has a long history of seeking help from the International Monetary Fund (IMF). Since the late 1980s, the country has gone to the IMF for help 13 times, most recently in 2019. Pakistan's economic problems include high levels of public-sector debt, a shortage of foreign exchange reserves required to service its international debt commitments, a persistent trade deficit, high inflation, and a weak currency. The Pakistan rupee has depreciated against the dollar from \$1 = 63 rupees in early 2008 to \$1 = 278 rupees by early March 2023. In its repeated trips to the IMF, Pakistan has sought financial assistance to protect its currency and boost its foreign exchange reserves. In return, the IMF has required that Pakistan take steps to stabilize its economy. Pakistan, however, has repeatedly failed to live up to its commitments under IMF agreements, primarily due to political opposition within the country, and its economic problems have persisted.

In 2019, Pakistan found itself confronting a familiar scenario—it was running out of foreign exchange reserves and was facing the prospect of defaulting on its international debt commitments. Almost all of Pakistan's foreign exchange reserves are held in U.S. dollars. The lack of foreign exchange reserves could be traced to two sources: (1) the country's persistent trade deficit, which effectively meant that dollars were flowing out of the country to pay for imports, and (2) repeated attempts by the Pakistani central bank to use its foreign exchange reserves to prop up the value of the rupee on the foreign exchange market. It did this by entering the market and buying rupees with U.S. dollars. The bank was doing this in an attempt to limit the inflationary consequences of depreciation in the value of the rupee.

In May 2019, the government of Pakistan, led by then–prime minister Imran Khan, reached an agreement to secure a \$6.5 billion loan from the IMF. In return, the IMF required the government to adopt a set of policies that were designed to restore domestic and external sustainability by reducing the public-sector deficit, reining in inflation, and letting the rupee depreciate on the foreign exchange markets. The policies include tax increases, higher interest rates, and less government spending, including a reduction in government subsidies designed to hold down the prices of gas and electricity. The loan was to be paid out over time in tranches, contingent upon Pakistan taking the actions it agreed to.

Unfortunately, as has occurred often in the past, Pakistan repeatedly failed to meet its commitments under the agreement. The IMF responded by holding up the release of loan tranches. Part of the problem is political chaos in Pakistan. Imran Khan was ousted in a parliament vote in 2022. There have been three different finance ministers since the loan was negotiated, and each has tried to renegotiate the loan to get a better deal for Pakistan—which, for Pakistan, means fewer commitments. To compound matters, in late 2022 the country was hit by devastating floods that left more than one-third of the country under water and caused an estimated \$30 billion in damage.

By early 2023, Pakistan was in dire need of unlocking \$1.1 billion in loans from the IMF to avoid possible default on its debt commitments. The country had less than \$3 billion in foreign exchange reserves left, barely enough to cover a month of imports. In extended negotiations between the government and the IMF, the IMF urged Pakistan to raise taxes on wealthy individuals and business that were doing well, while continuing to protect the poor (the World Bank estimates that over a third of the population lives below the poverty line). The IMF also required that subsidies be directed only toward the poor and that they should not benefit the wealthy and politically connected. The IMF also wanted the government stop trying to prop up the value of the rupee on the foreign exchange market.

In an effort to secure the loan, Pakistan has moved toward the IMF's position, letting the rupee fall against the dollar, cutting some subsidies, and raising interest rates to try and reduce inflation (which in early 2023 was running at a 50-year high of 28 percent). However, with parliamentary elections scheduled for August 2023, as it often has, the government resisted pressure from the IMF to move more aggressively on policy changes, primarily because it did not want to lose electoral support in advance of the elections.

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international monetary system

Institutional arrangements countries adopt to govern exchange rates.

floating exchange rate

A system under which the exchange rate for converting one currency into another is continuously adjusted depending on the laws of supply and demand.

Introduction

In this chapter, we look at the international monetary system and its role in determining exchange rates. The **international monetary system** refers to the institutional arrangements that govern exchange rates. In Chapter 10, we assumed the foreign exchange market was the primary institution for determining exchange rates, and the impersonal market forces of demand and supply determined the relative value of any two currencies (i.e., their exchange rate). Furthermore, we explained that the demand and supply of currencies are influenced by their respective countries' relative inflation rates and interest rates. When the foreign exchange market determines the relative value of a currency, we say that the country is adhering to a **floating exchange rate** regime. Four of the world's major trading currencies—the U.S.

pegged exchange rate

Currency value is fixed relative to a reference currency.

managed-float system

System under which some currencies are allowed to float freely, but the majority are either managed by government intervention or pegged to another currency.

dirty-float system

A system under which a country's currency is nominally allowed to float freely against other currencies but in which the government will intervene, buying and selling currency, if it believes that the currency has deviated too far from its fair value.

dollar, the European Union's euro, the Japanese yen, and the British pound—are all free to float against each other. Thus, market forces determine their exchange rates, which fluctuate against each other day to day, if not minute to minute. However, the exchange rates of many currencies are not determined by the free play of market forces; other institutional arrangements are adopted.

Many of the world's developing nations peg their currencies, primarily to the dollar or the euro. A **pegged exchange rate** means the value of the currency is fixed relative to a reference currency, such as the U.S. dollar, and then the exchange rate between that currency and other currencies is determined by the reference currency exchange rate.

Other countries, while not adopting a formal pegged rate, try to hold the value of their currency within some range against an important reference currency such as the U.S. dollar or a “basket” of currencies. This is often referred to as a **managed-float system** or a **dirty-float system**. It is a float because, in theory, the value of the currency is determined by market forces, but it is a managed (or dirty) float (as opposed to a clean float) because the central bank of a country will intervene in the foreign exchange market to try to maintain the value of its currency if it depreciates (or appreciates) too rapidly against an important reference currency. As suggested in the opening case, Pakistan has long operated with a managed-float system in which the country's central bank has often intervened in the foreign exchange market to prop up the value of the rupee against the U.S. dollar.



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The international monetary system captures our attention because here we are talking about the institutional arrangements that govern exchange rates, and this is a tricky business in which countries have leverage to influence their country's currency value but do not necessarily always use it. The options for how the value, or “rate,” is set for a currency are many: floating exchange rate, pegged exchange rate, dirty float, and fixed exchange rate are the ones covered in Chapter 11. We start the chapter with some history related to the “gold standard,” a practice that takes us back to ancient times. Technically, no country uses the gold standard any longer, but many, including the

United States, hold substantial gold reserves. The international monetary system depends on a lot of variables (see the globalEDGE™ Database of International Business Statistics, which we covered in Chapter 10). Today, these variables also include “behavioral” (perception) issues in addition to hard, concrete data. The globalEDGE™ Blog has been a favored “international business” vehicle to stay current on important topics, often related to monetary issues. Check out the globalEDGE™ Blog (globaledge.msu.edu/blog), see what is covered on monetary issues, and engage with people from around the world on issues that are of interest to you.

fixed exchange rate

A system under which the exchange rate for converting one currency into another is fixed.

European Monetary System (EMS)

EU system designed to create a zone of monetary stability in Europe, control inflation, and coordinate exchange rate policies of EU countries.

dollarization

The process of aligning a country's currency with the U.S. dollar.

Still other countries have operated with a **fixed exchange rate**, in which the values of a set of currencies are fixed against each other at some mutually agreed-on exchange rate. Before the introduction of the euro in 1999, several member states of the European Union operated with fixed exchange rates within the context of the **European Monetary System (EMS)**. For a quarter of a century after World War II, the world's major industrial nations participated in a fixed exchange rate system. Although this system collapsed in 1973, some people still argue that the world should attempt to reestablish it.

Another option is for a country to abandon its own currency and instead adopt another currency (typically the U.S. dollar—a process referred to as **dollarization**). Dollarization is sometimes used when a country is suffering from severe macroeconomic problems, such as high inflation, that are making its own currency worthless. This was the case with Ecuador in 2000, which abandoned its own currency in favor of the U.S. dollar after suffering from hyperinflation. Dollarization is now being considered as an option for Venezuela, a country whose currency has been rendered worthless by hyperinflation.

This chapter explains how the international monetary system works and points out its implications for international business. As we shall see, one of the interesting facets of the international monetary system is the central role played by the U.S. dollar. To understand how the system works, we must review its evolution. We begin with a discussion of the gold standard and

its breakup during the 1930s. Then, we discuss the 1944 Bretton Woods conference. The Bretton Woods conference created two major international institutions that play a role in the international monetary system—the International Monetary Fund (IMF) and the World Bank. The IMF was given the task of maintaining order in the international monetary system; the World Bank's role was to promote development. The Bretton Woods system of fixed exchange rates collapsed in 1973. Since then, the world has operated with a mixed system in which some currencies are allowed to float freely, but many are either managed by government intervention or pegged to another currency.

Notwithstanding the collapse of the fixed exchange rate system, today both the IMF and the World Bank continue to play major roles in the world economy and in the international monetary system. For example, as described in the opening case, the IMF repeatedly offered assistance to Pakistan to help that country deal with currency and debt crises. In 2019, the IMF stepped in to help Pakistan navigate its way through a currency crisis caused by decades of poor economic policies and expanding trade and budget deficits. Pakistan has not been able to generate enough foreign exchange reserves to cover its international obligations, including paying for imports and serving foreign debt. As its foreign reserves shrank, the Pakistan rupee has declined in value against major world currencies, boosting import prices and inflation. Typically, a loan from the IMF helps a country to serve its international obligations, but as in the Pakistan case, such loans come with conditions attached, requiring that the recipient take concrete steps to improve its economy. While the Pakistan government has followed some of the recommendations of the IMF, political conditions within Pakistan have made full implementation of IMF requirements fraught with challenges, and consequently, the Pakistan economy is still in a fragile state and may have to seek additional assistance in the future.

Finally, we discuss the implications of all this material for international business. We will see how the exchange rate policy adopted by a government can have an important impact on the outlook for business operations in a given country. We also look at how the policies adopted by the IMF can have an impact on the economic outlook for a country and, accordingly, on the costs and benefits of doing business in that country.

The Gold Standard



LO11-1

Describe the historical development of the modern global monetary system.

The gold standard had its origin in the use of gold coins as a medium of exchange, unit of account, and store of value—a practice that dates to ancient times. When international trade was limited in volume, payment for goods purchased from another country was typically made in gold or silver. However, as the volume of international trade expanded in the wake of the Industrial Revolution, a more convenient means of financing international trade was needed. Shipping large quantities of gold and silver around the world to finance international trade seemed impractical. The solution adopted was to arrange for payment in paper currency and for governments to agree to convert the paper currency into gold on demand at a fixed rate.

MECHANICS OF THE GOLD STANDARD

Pegging currencies to gold and guaranteeing convertibility is known as the **gold standard**. By 1880, most of the world's major trading nations—including Great Britain, Germany, Japan, and the United States—had adopted the gold standard. Given a common gold standard, the value of any currency in units of any other currency (the exchange rate) was easy to determine.

gold standard

The practice of pegging currencies to gold and guaranteeing convertibility.

gold par value

The amount of currency needed to purchase one ounce of gold.

For example, under the gold standard, 1 U.S. dollar was defined as equivalent to 23.22 grains of “fine” (pure) gold. Thus, one could, in theory, demand that the U.S. government convert that one dollar into 23.22 grains of gold. Because there are 480 grains in an ounce, one ounce of gold costs \$20.67 (480/23.22). The amount of a currency needed to purchase one ounce of gold was referred to as the **gold par value**. The British pound was valued at 113 grains of fine gold. In other words, one ounce of gold costs £4.25 (480/113). From the gold par values of pounds and dollars, we can calculate what the exchange rate was for converting pounds into dollars; it was $\text{£1} = \$4.87$ (i.e., $\$20.67/\text{£4.25}$).

It should be noted that while one could theoretically convert paper currency into gold, in practice central banks issued more currency than they had gold on hand. This practice, known as fractional reserve banking, is the cornerstone of modern banking practice. It was based on the expectation that at any one time, only a small number of currency holders would want their paper currency changed into gold, and thus the bank could create more money than it had gold.

STRENGTH OF THE GOLD STANDARD

balance-of-trade equilibrium

Reached when the income a nation’s residents earn from exports equals money paid for imports.

The great strength claimed for the gold standard was that it contained a powerful mechanism for achieving balance-of-trade equilibrium by all countries.¹ A country is said to be in **balance-of-trade equilibrium** when the income its residents earn from exports is equal to the money its residents pay to other countries for imports (the current account of its balance of payments is in balance). Suppose only two countries exist in the world, Japan and the United States. Imagine Japan’s trade balance is in surplus because it exports more to the United States than it imports from the United States. Japanese exporters are paid in U.S. dollars, which they exchange for Japanese yen at a Japanese bank. The Japanese bank submits the dollars to the U.S. government and demands payment of gold in return. (This is a simplification of what would occur, but it will make our point.)

Under the gold standard, when Japan has a trade surplus, a net flow of gold from the United States to Japan occurs. These gold flows automatically reduce the U.S. money supply and swell Japan’s money supply. As we saw in Chapter 10, a close connection exists between money supply growth and price inflation. An increase in money supply will raise prices in Japan, while a decrease in the U.S. money supply will push U.S. prices downward. The rise in the price of Japanese goods will decrease demand for these goods, while the fall in the price of U.S. goods will increase demand for these goods. Thus, Japan will start to buy more from the United States, and the United States will buy less from Japan, until a balance-of-trade equilibrium is achieved.

This adjustment mechanism seems so simple and attractive that even today, half a century after the final collapse of the gold standard, some people believe the world should return to a gold standard.

THE PERIOD BETWEEN THE WARS: 1918–1939

The gold standard worked reasonably well from the 1870s until the start of World War I in 1914, when it was abandoned. During the war, several governments financed part of their massive military expenditures by printing money. This resulted in inflation, and by the war’s end in 1918, price levels were higher everywhere. The United States returned to the gold standard in 1919, Great Britain in 1925, and France in 1928.

Great Britain returned to the gold standard by pegging the pound to gold at the prewar gold parity level of £4.25 per ounce, despite substantial inflation between 1914 and 1925. This priced British goods out of foreign markets, which pushed the country into a deep depression. When foreign holders of pounds lost confidence in Great Britain’s commitment to maintaining its currency’s value, they began converting their holdings of pounds into gold. The British government saw that it could not satisfy the demand for gold without seriously depleting its gold reserves, so it suspended convertibility in 1931.

The United States followed suit and left the gold standard in 1933 but returned to it in 1934, raising the dollar price of gold from \$20.67 per ounce to \$35.00 per ounce. Because more dollars were needed to buy an ounce of gold than before, the implication was that the dollar was worth less. This effectively amounted to a devaluation of the dollar relative to other currencies.

Thus, before the devaluation, the pound/dollar exchange rate was $\text{£}1 = \$4.87$, but after the devaluation it was $\text{£}1 = \$8.24$. By reducing the price of U.S. exports and increasing the price of imports, the government was trying to create employment in the United States by boosting output (the U.S. government was basically using the exchange rate as an instrument of trade policy—something it now accuses China of doing). However, a number of other countries adopted a similar tactic, and in the cycle of competitive devaluations that soon emerged, no country could win.

The net result was the shattering of any remaining confidence in the system. With countries devaluing their currencies at will, one could no longer be certain how much gold a currency could buy. Instead of holding onto another country's currency, people often tried to change it into gold immediately, lest the country devalue its currency in the intervening period. These actions put pressure on the gold reserves of various countries, forcing them to suspend gold convertibility and by the start of World War II in 1939, the gold standard was in tatters.



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The Bretton Woods System



LO11-2

Explain the role played by the World Bank and the IMF in the international monetary system.

In 1944, at the height of World War II, representatives from 44 countries met at Bretton Woods, New Hampshire, to design a new international monetary system. With the collapse of the gold standard and the Great Depression of the 1930s fresh in their minds, these leaders were determined to build an enduring economic order that would facilitate postwar economic growth. Participants had reached a consensus that fixed exchange rates were desirable. In addition, the conference participants wanted to avoid the senseless competitive devaluations of the 1930s, and they recognized that the gold standard would not ensure this. The major problem with the gold standard as previously constituted was that no multinational institution could stop countries from engaging in competitive devaluations.

The agreement reached at Bretton Woods established two multinational institutions—the International Monetary Fund (IMF) and the World Bank. The task of the IMF would be to maintain order in the international monetary system and that of the World Bank would be to promote general economic development. The Bretton Woods agreement also called for a system of fixed exchange rates that would be policed by the IMF. Under the agreement, all countries were to fix the value of their currency in terms of gold but were *not* required to exchange their currencies for gold. Only the dollar remained convertible into gold—at a price of \$35 per ounce. Each country decided what it wanted its exchange rate to be vis-à-vis the dollar and then calculated the gold par value of the currency based on that selected dollar exchange rate. All participating countries agreed to try to maintain the value of their currencies within 1 percent of the par value by buying or selling currencies (or gold) as needed. For example, if foreign exchange dealers were selling more of a country's currency than demanded, that country's government would intervene in the foreign exchange markets, buying its currency in an attempt to increase demand and maintain its gold par value.

Another aspect of the Bretton Woods agreement was a commitment not to use devaluation as a weapon of competitive trade policy. However, if a currency became too weak to defend, a devaluation of up to 10 percent would be allowed without any formal approval by the IMF. Larger devaluations required IMF approval.

THE ROLE OF THE IMF

The IMF Articles of Agreement were heavily influenced by the worldwide financial collapse, competitive devaluations, trade wars, high unemployment, hyperinflation in Germany and elsewhere, and general economic disintegration that occurred between the two World Wars. The aim of the Bretton Woods agreement, of which the IMF was the main custodian, was to try to avoid a repetition of that chaos through a combination of discipline and flexibility.

Discipline

A fixed exchange rate regime imposes discipline in two ways. First, the need to maintain a fixed exchange rate puts a brake on competitive devaluations and brings stability to the world trade environment. Second, a fixed exchange rate regime imposes monetary discipline on countries, thereby curtailing price inflation. For example, consider what would happen under a fixed exchange rate regime if Great Britain rapidly increased its money supply by printing pounds. As explained in Chapter 10, the increase in money supply would lead to price inflation. Given fixed exchange rates, inflation would make British goods uncompetitive in world markets, while the prices of imports would become more attractive in Great Britain. The result would be a widening trade deficit in Great Britain, with the country importing more than it exports. To correct this trade imbalance under a fixed exchange rate regime, Great Britain would be required to restrict the rate of growth in its money supply to bring price inflation back under control. Thus, fixed exchange rates are seen as a mechanism for controlling inflation and imposing economic discipline on countries.

Flexibility

Although monetary discipline was a central objective of the Bretton Woods agreement, participants recognized that a rigid policy of fixed exchange rates would be too inflexible and would probably break down just as the gold standard had. In some cases, a country's attempts to reduce its money supply growth and correct a persistent balance-of-payments deficit could force the country into recession and create high unemployment. The architects of the Bretton Woods agreement wanted to avoid high unemployment, so they built limited flexibility into the system. Two major features of the IMF Articles of Agreement fostered this flexibility: IMF lending facilities and adjustable parities.

The IMF stood ready to lend foreign currencies to members to tide them over during short periods of balance-of-payments deficits, when a rapid tightening of monetary or fiscal policy would hurt domestic employment. A pool of gold and currencies contributed by IMF members provided the resources for these lending operations. A persistent balance-of-payments deficit can lead to a depletion of a country's reserves of foreign currency, forcing it to devalue its currency. By providing deficit-laden countries with short-term foreign currency loans, IMF funds would buy time for countries to bring down their inflation rates and reduce their balance-of-payments deficits. The belief was that such loans would reduce pressures for devaluation and allow for a more orderly and less painful adjustment.

Countries were to be allowed to borrow a limited amount from the IMF without adhering to any specific agreements. However, extensive drawings from IMF funds would require a country to agree to increasingly stringent IMF supervision of its macroeconomic policies. Heavy borrowers from the IMF must agree to monetary and fiscal conditions set down by the IMF, which typically included IMF-mandated targets on domestic money supply growth, exchange rate policy, tax policy, government spending, and so on.

The system of adjustable parities allowed for the devaluation of a country's currency by more than 10 percent if the IMF agreed that a country's balance of payments was in "fundamental disequilibrium." The term *fundamental disequilibrium* was not defined in the IMF's Articles of Agreement, but it was intended to apply to countries that had suffered permanent adverse shifts

in the demand for their products. Without devaluation, such a country would experience high unemployment and a persistent trade deficit until the domestic price level had fallen far enough to restore a balance-of-payments equilibrium. The belief was that devaluation could help sidestep a painful adjustment process in such circumstances.

THE ROLE OF THE WORLD BANK

The official name for the World Bank is the International Bank for Reconstruction and Development (IBRD). When the Bretton Woods participants established the World Bank, the need to reconstruct the war-torn economies of Europe was foremost in their minds. The bank's initial mission was to help finance the building of Europe's economy by providing low-interest loans. As it turned out, the World Bank was overshadowed in this role by the Marshall Plan, under which the United States lent money directly to European nations to help them rebuild. So the bank turned its attention to development and began lending money to third-world nations. During the 1950s, the bank concentrated on public-sector projects. Power stations, road building, and other transportation investments were much in favor. During the 1960s, the bank also began to lend heavily in support of agriculture, education, population control, and urban development.

The bank lends money under two schemes. Under the IBRD scheme, money is raised through bond sales in the international capital market. Borrowers pay what the bank calls a market rate of interest—the bank's cost of funds plus a margin for expenses. This “market” rate is lower than commercial banks' market rate. Under the IBRD scheme, the bank offers low-interest loans to risky customers whose credit rating is often poor, such as the governments of underdeveloped nations.

A second scheme is overseen by the International Development Association (IDA), an arm of the bank created in 1960. Resources to fund IDA loans are raised through subscriptions from wealthy members such as the United States, Japan, and Germany. IDA loans go only to the poorest countries. Borrowers have up to 50 years to repay at an interest rate of less than 1 percent a year. The world's poorest nations receive grants and interest-free loans.



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The Collapse of the Fixed Exchange Rate System



LO11-3

Compare and contrast the differences between a fixed and a floating exchange rate system.

The system of fixed exchange rates established at Bretton Woods worked well until the late 1960s, when it began to show signs of strain. The system finally collapsed in 1973, and since then, we have had a managed-float system. To understand why the system collapsed, one must appreciate the special role of the U.S. dollar in the system. As the only currency that could be converted into gold and the currency that served as the reference point for all others, the dollar

occupied a central place in the system. Any pressure on the dollar to devalue could wreak havoc with the system, and that is what occurred.

Most economists trace the breakup of the fixed exchange rate system to the U.S. macroeconomic policy package of 1965–1968.² To finance both the Vietnam conflict and his welfare programs, President Lyndon Johnson backed an increase in U.S. government spending that was not financed by an increase in taxes. Instead, it was financed by an increase in the money supply, which led to a rise in price inflation from less than 4 percent in 1966 to close to 9 percent by 1968. At the same time, the rise in government spending had stimulated the economy. With more money in their pockets, people spent more—particularly on imports—and the U.S. trade balance began to deteriorate.

The increase in inflation and the worsening of the U.S. foreign trade position gave rise to speculation in the foreign exchange market that the dollar would be devalued. Things came to a head in spring 1971, when U.S. trade figures showed that for the first time since 1945, the United States was importing more than it was exporting. This set off massive purchases of German deutsche marks in the foreign exchange market by speculators who guessed that the mark would be revalued against the dollar. On a single day, May 4, 1971, the Bundesbank (Germany's central bank) had to buy \$1 billion to hold the dollar/deutsche mark exchange rate at its fixed exchange rate, given the great demand for deutsche marks. On the morning of May 5, the Bundesbank purchased another \$1 billion during the first hour of foreign exchange trading! At that point, the Bundesbank faced the inevitable and allowed its currency to float.

In the weeks following the decision to float the deutsche mark, the foreign exchange market became increasingly convinced that the dollar would have to be devalued. However, devaluation of the dollar was no easy matter. Under the Bretton Woods provisions, any other country could change its exchange rates against all currencies simply by fixing its dollar rate at a new level. But as the key currency in the system, the dollar could be devalued only if all countries agreed to simultaneously revalue against the dollar. Many countries did not want to do this because it would make their products more expensive relative to U.S. products.

To force the issue, President Richard Nixon announced in August 1971 that the dollar was no longer convertible into gold. He also announced that a new 10 percent tax on imports would remain in effect until U.S. trading partners agreed to revalue their currencies against the dollar. These actions brought the trading partners to the bargaining table, and in December 1971, an agreement was reached to devalue the dollar by about 8 percent against foreign currencies. The import tax was then removed. The problem was not solved, however. The U.S. balance-of-payments position continued to deteriorate throughout 1973, while the nation's money supply continued to expand at an inflationary rate. Speculation continued to grow that the dollar was still overvalued and that a second devaluation would be necessary. In anticipation, foreign exchange dealers began converting dollars to deutsche marks and other currencies. After a massive wave of speculation in February 1973, which culminated with European central banks spending \$3.6 billion on March 1 to try to prevent their currencies from appreciating against the dollar, the foreign exchange market was closed. When the foreign exchange market reopened March 19, the currencies of Japan and most European countries were floating against the dollar, although many developing countries continued to peg their currency to the dollar—and many do so to this day. At that time, the switch to a floating system was viewed as a temporary response to unmanageable speculation in the foreign exchange market. But 50 years have now passed since the Bretton Woods system of fixed exchange rates collapsed, and the temporary solution looks permanent.

The Bretton Woods system had an Achilles' heel: The system could not work if its key currency, the U.S. dollar, was under speculative attack. The Bretton Woods system could work only as long as the U.S. inflation rate remained low and the United States did not run a balance-of-payments deficit. Once these things occurred, the system soon became strained to the breaking point.

The Floating Exchange Rate Regime

The floating exchange rate regime that followed the collapse of the fixed exchange rate system was formalized in January 1976, when IMF members met in Jamaica and agreed to the rules for the international monetary system that are in place today.

THE JAMAICA AGREEMENT

The Jamaica meeting revised the IMF's Articles of Agreement to reflect the new reality of floating exchange rates. The main elements of the Jamaica agreement include the following:

- Floating rates were declared acceptable. IMF members were permitted to enter the foreign exchange market to even out “unwarranted” speculative fluctuations.
- Gold was abandoned as a reserve asset. The IMF returned its gold reserves to members at the current market price, placing the proceeds in a trust fund to help poor nations. IMF members were permitted to sell their own gold reserves at the market price.
- Total annual IMF quotas—the amount member countries contribute to the IMF—were increased to \$41 billion. (Since then, they have been increased to \$767 billion, while the membership of the IMF has been expanded to include 190 countries. Non-oil-exporting, less-developed countries were given greater access to IMF funds.)

EXCHANGE RATES SINCE 1973

Since March 1973, exchange rates have become much more volatile and less predictable than they were between 1945 and 1973.³ This volatility has been partially due to a number of unexpected shocks to the world monetary system, including:

- The oil crisis in 1971, when the Organization of the Petroleum Exporting Countries (OPEC) quadrupled the price of oil. The harmful effect of this on the U.S. inflation rate and trade position resulted in a further decline in the value of the dollar.
- The loss of confidence in the dollar that followed a sharp rise in the U.S. inflation rate in 1977–1978.
- The oil crisis of 1979, when OPEC once again increased the price of oil dramatically—this time, doubling it.
- The unexpected rise in the dollar between 1980 and 1985, despite a deteriorating balance-of-payments picture.
- The rapid fall of the U.S. dollar against the Japanese yen and German deutsche mark between 1985 and 1987 and against the yen between 1993 and 1995.
- The partial collapse of the European Monetary System in 1992.
- The 1997 Asian currency crisis, when the Asian currencies of several countries—including South Korea, Indonesia, Malaysia, and Thailand—lost between 50 and 80 percent of their value against the U.S. dollar within a few months.
- The global financial crisis of 2008–2010 and the sovereign debt crisis in the European Union during 2010–2011.

Figure 11.1 summarizes how the value of the U.S. dollar has fluctuated against an index of trading currencies between 1973 and 2019. The index, which was set equal to 100 in March 1973, is a weighted average of the foreign exchange values of the U.S. dollar against a basket of other currencies. The definition of the index changed in February 2019, and the figures since then are not strictly comparable with those prior to 2019, so they are not reported here. However, we should note that the dollar strengthened against most major currencies in the 2020–2022 period, as investors responded to the COVID pandemic and geopolitical events (including, most notably, the Russian invasion of Ukraine) by investing more in dollar-denominated U.S. Treasury bills, which were seen as a safe asset in troubled times.

We are focusing on the U.S. dollar because it plays such a pivotal role in the international monetary system. Around 60 percent of global foreign exchange reserves are held in U.S. dollars, close to 90 percent of all foreign exchange transactions involve U.S. dollars, 40 percent of the world's debt is issued in dollars, and the dollar is used in 40 percent of international payments. The dollar enjoys this role because the United States remains the world's largest economy, foreign investors have faith in the U.S. Federal Reserve Board to control inflation and maintain the value of the dollar, and dollar-denominated assets can be freely exchanged in a very liquid market.



FIGURE 11.1 Major currencies dollar index, 1973–2019.

Source: Data from www.federalreserve.gov.

An interesting phenomenon in Figure 11.1 is the rapid rise in the value of the dollar between 1980 and 1985 and its subsequent fall between 1985 and 1988. A similar, though less pronounced, rise and fall in the value of the dollar occurred between 1995 and 2012. You may also notice an uptick in the value of the dollar between mid-2014 and early 2019. We briefly discuss the rise and fall of the dollar during these periods, because this tells us something about how the international monetary system has operated in recent years.⁴

The rise in the value of the dollar between 1980 and 1985 occurred when the United States was running a large and growing trade deficit, importing substantially more than it exported. Conventional wisdom would suggest that the increased supply of dollars in the foreign exchange market as a result of the trade deficit should lead to a reduction in the value of the dollar, but as shown in Figure 11.1, it increased in value. Why?

A number of favorable factors overcame the unfavorable effect of a trade deficit. Strong economic growth in the United States attracted heavy inflows of capital from foreign investors seeking high returns on capital assets. High real interest rates attracted foreign investors seeking high returns on financial assets. At the same time, political turmoil in other parts of the world, along with relatively slow economic growth in the developed countries of Europe, helped create the view that the United States was a good place to invest. These inflows of capital increased the demand for dollars in the foreign exchange market, which pushed the value of the dollar upward against other currencies.

The fall in the value of the dollar between 1985 and 1988 was caused by a combination of government intervention and market forces. The rise in the dollar, which priced U.S. goods out of foreign markets and made imports relatively cheap, had contributed to a dismal trade picture. In 1985, the United States posted a then-record-high trade deficit of more than \$160 billion. This trade deficit led to growth in demands for protectionism in the United States. In September 1985, the finance ministers and central bank governors of the so-called Group of Five major industrial countries (Great Britain, France, Japan, Germany, and the United States) met at the Plaza Hotel in New York City and reached what was later referred to as the Plaza Accord. They announced that it would be desirable for most major currencies to appreciate in relation to the U.S. dollar and pledged to intervene in the foreign exchange markets, selling dollars, to encourage this objective. The dollar had already begun to weaken during summer 1985, and this announcement further accelerated the decline.

The dollar continued to decline until 1987. The governments of the Group of Five then began to worry that the dollar might decline too far, so the finance ministers of the Group of Five met in Paris in February 1987 and reached a new agreement known as the Louvre Accord. They agreed that exchange rates had been realigned sufficiently and pledged to support the stability of exchange rates around their current levels by intervening in the foreign exchange markets when

necessary to buy and sell currency. Although the dollar continued to decline for a few months after the Louvre Accord, the rate of decline slowed, and by early 1988, the decline had ended.

Except for a brief speculative flurry around the time of the Persian Gulf War in 1991, the dollar was relatively stable for the first half of the 1990s. However, in the late 1990s, the dollar again began to appreciate against most major currencies, including the euro after its introduction, even though the United States was still running a significant balance-of-payments deficit. Once again, the driving force for the appreciation in the value of the dollar was that foreigners continued to invest in U.S. financial assets, primarily stocks and bonds, and the inflow of money drove up the value of the dollar on foreign exchange markets. The inward investment was due to a belief that U.S. financial assets offered a favorable rate of return.

By 2002, foreigners had started to lose their appetite for U.S. stocks and bonds, and the inflow of money into the United States slowed. Instead of reinvesting dollars earned from exports to the United States in U.S. financial assets, they exchanged those dollars for other currencies, particularly euros, to invest them in non-dollar-denominated assets. One reason for this was the continued growth in the U.S. trade deficit, which hit a record \$791 billion in 2005 (by 2016, it had fallen to \$502 billion). Although U.S. trade deficits had been setting records for decades, this deficit was the largest ever when measured as a percentage of the country's GDP (6.3 percent of GDP in 2005).

The record deficit meant that even more dollars were flowing out of the United States into foreign hands, and those foreigners were less inclined to reinvest those dollars in the United States at a rate required to keep the dollar stable. This growing reluctance of foreigners to invest in the United States was in turn due to several factors. First, U.S. economic activity slowed down during 2001–2002. Second, the U.S. government's budget deficit expanded rapidly after 2001. This led to fears that ultimately the budget deficit would be financed by an expansionary monetary policy that could lead to higher price inflation. Third, from 2003 onward, U.S. government officials began to “talk down” the value of the dollar, in part because the administration believed that a cheaper dollar would increase exports and reduce imports, thereby improving the U.S. balance-of-trade position.⁵ Foreigners saw this as a signal that the U.S. government would not intervene in the foreign exchange markets to prop up the value of the dollar, which increased their reluctance to reinvest dollars earned from export sales in U.S. financial assets. As a result of these factors, demand for dollars weakened, and the value of the dollar slid on the foreign exchange markets—hitting an index value of 80.5 in June 2011, the lowest value since the index began in 1973. Some believed that the dollar would have fallen even further had not oil-producing states recycled the dollars they were earning from sales of crude oil back into the U.S. economy. At the time, these states were benefiting from high oil prices (oil is priced in U.S. dollars), and they chose to invest the dollars they earned back into the United States, rather than selling them for another currency.

Interestingly, from mid-2008 through early 2009, the dollar staged a moderate rally against major currencies, despite the fact that the U.S. economy was suffering from a serious financial crisis. The reason seems to be that despite the United States' problems, things were even worse in many other countries, and foreign investors saw the dollar as a safe haven and put their money in low-risk U.S. assets, particularly low-yielding U.S. government bonds. This rally faltered in mid-2009 as investors became worried about the level of U.S. indebtedness. However, between 2014 and early 2021, the dollar yet again increased significantly in value, primarily because of the strength of the U.S. economy, which had emerged from the great financial crisis of 2008–2009 in better shape than any other major developed nation, with higher economic growth rates and lower levels of unemployment.



Did You Know?

Did you know that the Turkish lira lost 40 percent of its value against the U.S. dollar during the first nine months of 2018? Visit your instructor's Connect® course and click on your eBook or SmartBook® to view a short video explanation from the author.

This review tells us that in recent history, both market forces and government intervention have determined the value of the dollar. Under a floating exchange rate regime, market forces have produced a volatile dollar exchange rate. Governments have sometimes responded by intervening in the market—buying and selling dollars—in an attempt to limit the market’s volatility and to correct what they see as overvaluation (in 1985) or potential undervaluation (in 1987) of the dollar. In addition to direct intervention, statements from government officials have frequently influenced the value of the dollar. The dollar may not have declined by as much as it did in 2004, for example, had not U.S. government officials publicly ruled out any action to stop the decline. Paradoxically, a signal not to intervene can affect the market. The frequency of government intervention in the foreign exchange market explains why the current system is sometimes thought of as a *managed-float system* or a *dirty-float system*.



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Fixed versus Floating Exchange Rates

The breakdown of the Bretton Woods system has not stopped the debate about the relative merits of fixed versus floating exchange rate regimes. Disappointment with the system of floating rates in recent years has led to renewed debate about the merits of fixed exchange rates. This section reviews the arguments for fixed and floating exchange rate regimes.⁶ We discuss the case for floating rates before studying why many critics are disappointed with the experience under floating exchange rates and yearn for a system of fixed rates.

THE CASE FOR FLOATING EXCHANGE RATES

The case in support of floating exchange rates has three main elements: monetary policy autonomy, automatic trade balance adjustments, and economic recovery following a severe economic crisis.

Monetary Policy Autonomy

It is argued that under a fixed system, a country’s ability to expand or contract its money supply as it sees fit is limited by the need to maintain exchange rate parity. Monetary expansion can lead to inflation, which puts downward pressure on a fixed exchange rate (as predicted by the PPP theory; see Chapter 10). Similarly, monetary contraction requires high interest rates (to reduce the demand for money). Higher interest rates lead to an inflow of money from abroad, which puts upward pressure on a fixed exchange rate. Thus, to maintain exchange rate parity under a fixed system, countries were limited in their ability to use monetary policy to expand or contract their economies.

Advocates of a floating exchange rate regime argue that removal of the obligation to maintain exchange rate parity would restore monetary control to a government. If a government faced with unemployment wanted to increase its money supply to stimulate domestic demand and reduce unemployment, it could do so unencumbered by the need to maintain its exchange rate. While monetary expansion might lead to inflation, this would lead to a depreciation in the country’s currency. If PPP theory is correct, the resulting currency depreciation on the foreign exchange markets should offset the effects of inflation. Although under a floating exchange rate regime, domestic inflation would have an impact on the exchange rate, it should have no impact on businesses’ international cost competitiveness due to exchange rate depreciation. The rise in domestic costs should be exactly offset by the fall in the value of the country’s currency on the foreign exchange markets. Similarly, a government could use monetary policy to contract the economy without worrying about the need to maintain parity.

Trade Balance Adjustments

Under the Bretton Woods system, if a country developed a permanent deficit in its balance of trade (importing more than it exported) that could not be corrected by domestic policy, this would require the IMF to agree to currency devaluation. Critics of this system argue that the adjustment mechanism works much more smoothly under a floating exchange rate regime. They argue that if a country is running a trade deficit, the imbalance between the supply and demand of that country's currency in the foreign exchange markets (supply exceeding demand) will lead to depreciation in its exchange rate. In turn, by making its exports cheaper and its imports more expensive, exchange rate depreciation should correct the trade deficit.

Crisis Recovery

Advocates of floating exchange rates also argue that exchange rate adjustments can help a country to deal with economic crises. When a country is hit by a severe economic crisis, its currency typically declines on foreign exchange markets. The reason for this is that investors respond to the crisis by taking their money out of the country, selling the local currency, and driving down its value. At some point, however, the currency becomes so cheap that it starts to stimulate exports. This is what occurred in Iceland after the krona lost 50 percent of its value against the U.S. dollar and euro following a banking crisis in 2008. By 2009, exports of fish and aluminum from Iceland were booming, which helped pull the Icelandic economy out of a recession. A similar process occurred in South Korea after the 1997 Asian banking crisis. The value of the South Korean won plunged to 1,700 per dollar from around 800. In turn, the cheap won helped South Korea increase its exports and resulted in an export-led economic recovery. On the other hand, in both countries, the declining value of the currency did raise import prices and led to an increase in inflation, so a price has to be paid for an export-led recovery due to falling currency values.

A contrast can be drawn with the recent situation in Greece, where the economy imploded following the 2008–2009 global financial crisis and has struggled to recover. Part of the problem in Greece is that it gave up its own currency to adopt the euro in 2001, and the euro has remained quite strong; thus, Greece cannot rely on a falling local currency to boost exports and stimulate economic recovery.

THE CASE FOR FIXED EXCHANGE RATES

The case for fixed exchange rates rests on arguments about monetary discipline, speculation, uncertainty, and the lack of connection between the trade balance and exchange rates.

Monetary Discipline

We have already discussed the nature of monetary discipline inherent in a fixed exchange rate system when we discussed the Bretton Woods system. The need to maintain fixed exchange rate parity ensures that governments do not expand their money supplies at inflationary rates. While advocates of floating rates argue that each country should be allowed to choose its own inflation rate (the monetary autonomy argument), advocates of fixed rates argue that governments all too often give in to political pressures and expand the monetary supply far too rapidly, causing unacceptably high price inflation. A fixed exchange rate regime would ensure that this does not occur.

Speculation

Critics of a floating exchange rate regime also argue that speculation can cause fluctuations in exchange rates. They point to the dollar's rapid rise and fall during the 1980s, which they claim had nothing to do with comparative inflation rates and the U.S. trade deficit but everything to do with speculation. They argue that when foreign exchange dealers see a currency depreciating, they tend to sell the currency in the expectation of future depreciation, regardless of the currency's longer-term prospects. As more traders jump on the bandwagon, the expectations of depreciation are realized. Such destabilizing speculation tends to accentuate the fluctuations around the exchange rate's long-run value. It can damage a country's economy by distorting export and

import prices. Thus, advocates of a fixed exchange rate regime argue that such a system will limit the destabilizing effects of speculation.

Uncertainty

Speculation also adds to the uncertainty surrounding future currency movements that characterizes floating exchange rate regimes. The unpredictability of exchange rate movements in the post-Bretton Woods era has made business planning difficult, and it adds risk to exporting, importing, and foreign investment activities. Given a volatile exchange rate, international businesses do not know how to react to the changes—so often they do not react. Why change plans for exporting, importing, or foreign investment after a 6 percent fall in the dollar this month when the dollar may rise 6 percent next month? This uncertainty, according to the critics, dampens the growth of international trade and investment. They argue that a fixed exchange rate, by eliminating such uncertainty, promotes the growth of international trade and investment. Advocates of a floating system reply that the forward exchange market ensures against the risks associated with exchange rate fluctuations (see Chapter 10), so the adverse impact of uncertainty on the growth of international trade and investment has been overstated.

Trade Balance Adjustments and Economic Recovery

Those in favor of floating exchange rates argue that floating rates help adjust trade imbalances and can assist with economic recovery after a crisis. Critics question the closeness of the link between the exchange rate, the trade balance, and economic growth. They claim that trade deficits are determined by the balance between savings and investment in a country, not by the external value of its currency.⁷ They argue that depreciation in a currency will lead to inflation (due to the resulting increase in import prices). This inflation, they state, will wipe out any apparent gains in cost competitiveness that arise from currency depreciation. In other words, a depreciating exchange rate will not boost exports and reduce imports, as advocates of floating rates claim; it will simply boost price inflation. In support of this argument, those who favor fixed rates point out that the 40 percent drop in the value of the dollar between 1985 and 1988 did not correct the U.S. trade deficit. In reply, advocates of a floating exchange rate regime argue that between 1985 and 1992, the U.S. trade deficit fell from more than \$160 billion to about \$70 billion, and they attribute this in part to the decline in the value of the dollar. Moreover, the experience of countries like South Korea and Iceland seems to suggest that floating rates can help a country recover from a severe economic crisis.

WHO IS RIGHT?

Which side is right in the vigorous debate between those who favor a fixed exchange rate and those who favor a floating exchange rate? Economists cannot agree. Business, as a major player on the international trade and investment scene, has a large stake in the resolution of the debate. Would international business be better off under a fixed regime, or are flexible rates better? The evidence is not clear.

However, a fixed exchange rate regime modeled along the lines of the Bretton Woods system probably will not work. Speculation ultimately broke the system, a phenomenon that advocates of fixed rate regimes claim is associated with floating exchange rates! Nevertheless, a different kind of fixed exchange rate system might be more enduring and might foster the stability that would facilitate more rapid growth in international trade and investment. In the next section, we look at potential models for such a system and the problems with such systems.



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Exchange Rate Regimes in Practice



LO11-4

Identify exchange rate regimes used in the world today and why countries adopt different exchange rate regimes.

Governments around the world pursue a number of different exchange rate policies. These range from a pure “free float” in which the exchange rate is determined by market forces, to a pegged system that has some aspects of the pre-1973 Bretton Woods system of fixed exchange rates. Some 21 percent of the IMF’s members allow their currency to float freely. Another 23 percent intervene in only a limited way (the so-called managed float as practiced by China, among other nations—see the accompanying Country Focus for details). A further 5 percent of IMF members now have no separate legal tender of their own (this figure excludes the European Union countries that have adopted the euro). These are typically smaller states, mostly in Africa or the Caribbean, that have no domestic currency and have adopted a foreign currency as legal tender within their borders, typically the U.S. dollar or the euro (see the opening case). The remaining countries use more inflexible systems, including a fixed peg arrangement (43 percent) under which they peg their currencies to other currencies, such as the U.S. dollar or the euro, or to a basket of currencies. Other countries have adopted a system under which their exchange rate is allowed to fluctuate against other currencies within a target zone (an adjustable peg system). In this section, we look more closely at the mechanics and implications of exchange rate regimes that rely on a currency peg or target zone.

COUNTRY FOCUS

China’s Exchange Rate Regime

For years, some politicians in the United States have claimed that the Chinese actively manipulate their currency, the yuan, keeping its value low against the dollar and other major currencies in order to boost Chinese exports. In November 2015, for example, presidential hopeful Donald Trump claimed that “the wanton manipulation of China’s currency” is “robbing Americans of billions of dollars in capital and millions of jobs.” But is this claim true? Would it even be possible for China to manipulate the foreign exchange markets to artificially depress the value of its currency? To answer these questions, one needs to look at the history of exchange rate determination for China and understand something about how the international monetary system actually works.

For most of its history, the Chinese yuan was pegged to the U.S. dollar at a fixed exchange rate. When China started to open up its economy to foreign trade and investment in the 1980s, the yuan was devalued by the Chinese government in order to improve the competitiveness of Chinese exports. Thus, the official yuan/USD pegged exchange rate was increased from 1.50 yuan per U.S. dollar in

1980 to 8.62 yuan per U.S. dollar in 1994. With China’s exports growing and the country running a growing current account trade surplus, pressure began to increase for China to let its currency appreciate. In response, between 1997 and 2005, the exchange rate was fixed at 8.27 yuan per U.S. dollar, which represented a small appreciation. One could argue that during this period, China’s currency was indeed undervalued and that this was the result of government policy.

By the 2000s, China’s growing importance in the global economy and the rise of its export-led economy led to calls for the country to reevaluate its fixed exchange rate policy. In response, in July 2005, the country adopted a managed floating exchange rate system. Under this system, the exchange rate for the yuan was set with reference to a basket of foreign currencies that included the U.S. dollar, the euro, the Japanese yen, and the British pound. The daily exchange rate was allowed to float within a narrow band of 0.3 percent around the central parity. The daily band was extended to 0.5 percent in 2007, 1 percent in 2012, and 2 percent in 2014.

Over time, this managed-float system allowed for the appreciation of the Chinese yuan. For example, against the U.S. dollar, the exchange rate changed from 8.27 yuan per dollar in mid-2005 to 6.0875 yuan per U.S. dollar on

July 20, 2015, representing an appreciation of 26 percent. More generally, the effective exchange rate index of the yuan against a basket of more than 60 other currencies increased from 86.3 in July 2005 to 123.8 by early 2016, representing an appreciation of 43 percent. The yuan has appreciated by less than this against the U.S. dollar primarily because the U.S. dollar has also been relatively strong and appreciated against many other currencies over the same time period.

These data suggest that rather than artificially trying to keep its currency undervalued, since July 2005, the Chinese have allowed the yuan to increase in value against other currencies, albeit within the constraints imposed by the managed float. In late 2015, this commitment was put to the test when a slowdown in the rate of growth of the Chinese economy led to an outflow of capital from China, which put downward pressure on the yuan. The Chinese responded by trying to maintain the value of the yuan, using its foreign exchange reserves, which are primarily held in U.S. dollars, to buy yuan on the open market and shore up its value. Reports suggest that China spent \$500 billion in 2015 to shore up the value of the yuan and more than \$1 trillion in 2016. These actions reduced China's foreign exchange reserves to \$3.011 trillion by January

2017, the lowest level since 2012. China appears to be trying to keep the yuan from depreciating below 7 yuan to the U.S. dollar (in March 2023, it was trading at about 6.9 yuan to the dollar).

One reason for China to protect the value of the yuan against the dollar is that a large number of Chinese companies have dollar-denominated debt. If the yuan falls against the dollar, the price of serving that debt goes up when translated into yuan. This could stress the financials of those companies (possibly pushing some into bankruptcy) and make it more difficult for China to hit the government's economic growth targets. Another reason is that China might want to head off charges from American politicians that it continues to keep the value of its currency artificially low to boost exports.

Sources: T. Hult, "The U.S. Shouldn't Fret over Cheaper Yuan," *Time*, August 14, 2015; "The Yuan and the Markets," *The Economist*, January 16, 2016; M. Gesiotto, "The Negative Effects of China's Currency Manipulation Explained," *The Washington Times*, November 13, 2015; M. Slaughter, "The Myths of China's Currency Manipulation," *The Wall Street Journal*, January 8, 2016; "The Curious Case of China's Currency," *The Economist*, August 11, 2015; L. Wei, "China Foreign Exchange Reserves Keep Dropping," *The Wall Street Journal*, January 8, 2017.

PEGGED EXCHANGE RATES

Under a pegged exchange rate regime, a country will peg the value of its currency to that of a major currency so that, for example, as the U.S. dollar rises in value, its own currency rises too. Pegged exchange rates are popular among many of the world's smaller nations. As with a full fixed exchange rate regime, the great virtue claimed for a pegged exchange rate is that it imposes monetary discipline on a country and leads to low inflation. For example, if Belize pegs the value of the Belizean dollar to that of the U.S. dollar so that $\text{US\$1} = \text{B\$1.97}$, then the Belizean government must make sure the inflation rate in Belize is similar to that in the United States. If the Belizean inflation rate is greater than the U.S. inflation rate, this will lead to pressure to devalue the Belizean dollar (i.e., to alter the peg). To maintain the peg, the Belizean government would be required to rein in inflation. Of course, for a pegged exchange rate to impose monetary discipline on a country, the country whose currency is chosen for the peg must also pursue sound monetary policy.

Evidence shows that adopting a pegged exchange rate regime moderates inflationary pressures in a country. An IMF study concluded that countries with pegged exchange rates had an average annual inflation rate of 8 percent, compared with 14 percent for intermediate regimes and 16 percent for floating regimes.⁸ However, many countries operate with only a nominal peg and in practice are willing to devalue their currency rather than pursue a tight monetary policy. A smaller country can find it very difficult to maintain a peg against another currency if capital is flowing out of the country and foreign exchange traders are speculating against the currency. Something like this occurred in 1997 when a combination of adverse capital flows and currency speculation forced several Asian countries, including Thailand and Malaysia, to abandon pegs against the U.S. dollar and let their currencies float freely. Malaysia and Thailand would not have been in this position had they dealt with a number of problems that began to arise in their economies during the 1990s, including excessive private-sector debt and expanding current account trade deficits.

CURRENCY BOARDS

Hong Kong's experience during the 1997 Asian currency crisis added a new dimension to the debate over how to manage a pegged exchange rate. During late 1997, when other Asian currencies were collapsing, Hong Kong maintained the value of its currency against the U.S. dollar at about $\text{US\$1} = \text{HK\$7.80}$ despite several concerted speculative attacks. Hong Kong's currency

board has been given credit for this success. A country that introduces a **currency board** commits itself to converting its domestic currency on demand into another currency at a fixed exchange rate. To make this commitment credible, the currency board holds reserves of foreign currency equal at the fixed exchange rate to at least 100 percent of the domestic currency issued. The system used in Hong Kong means its currency must be fully backed by the U.S. dollar at the specified exchange rate. This is still not a true fixed exchange rate regime because the U.S. dollar—and, by extension, the Hong Kong dollar—floats against other currencies, but it has some features of a fixed exchange rate regime.

Under this arrangement, the currency board can issue additional domestic notes and coins only when there are foreign exchange reserves to back it. This limits the ability of the government to print money and, thereby, create inflationary pressures. Under a strict currency board system, interest rates adjust automatically. If investors want to switch out of domestic currency into, for example, U.S. dollars, the supply of domestic currency will shrink. This will cause interest rates to rise until holding the local currency eventually becomes attractive for investors again. In the case of Hong Kong, the interest rate on three-month deposits climbed as high as 20 percent in late 1997, as investors switched out of Hong Kong dollars and into U.S. dollars. The dollar peg held, however, and interest rates declined again.

Since its establishment in 1983, the Hong Kong currency board has weathered several storms. This success persuaded several other countries in the developing world to consider a similar system. Argentina introduced a currency board in 1991 (but abandoned it in 2002), and Bulgaria, Estonia, and Lithuania have all gone down this road in recent years. Despite interest in the arrangement, however, critics are quick to point out that currency boards have their drawbacks.⁹ If local inflation rates remain higher than the inflation rate in the country to which the currency is pegged, the currencies of countries with currency boards can become noncompetitive and overvalued (this is what happened in the case of Argentina, when it had a currency board). Also, under a currency board system, the government lacks the ability to set interest rates. Interest rates in Hong Kong, for example, are effectively set by the U.S. Federal Reserve. In addition, economic collapse in Argentina in 2001 and the subsequent decision to abandon its currency board dampened much of the enthusiasm for this mechanism of managing exchange rates.

currency board

Means of controlling a country's currency.



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Crisis Management by the IMF



LO11-5

Understand the debate surrounding the role of the IMF in the management of financial crises.

Many observers initially believed that the collapse of the Bretton Woods system in 1973 would diminish the role of the IMF within the international monetary system. The IMF's original function was to provide a pool of money from which members could borrow, short term, to adjust their balance-of-payments position and maintain their exchange rate. Some believed the demand for short-term loans would be considerably diminished under a floating exchange rate regime. A trade deficit would presumably lead to a decline in a country's exchange rate, which would help reduce imports and boost exports. No temporary IMF adjustment loan would be needed. Consistent with this, after 1973, most industrialized countries tended to let the foreign exchange

market determine exchange rates in response to demand and supply. Since the early 1970s, the rapid development of global capital markets has generally allowed developed countries such as Great Britain and the United States to finance their deficits by borrowing private money, as opposed to drawing on IMF funds.

Despite these developments, the activities of the IMF have expanded over the past 30 years. By 2023, the IMF had 190 members, around 40 of which had some kind of IMF program in place. In 1997, the institution implemented its largest rescue packages until that date, committing more than \$110 billion in short-term loans to three troubled Asian countries—South Korea, Indonesia, and Thailand. This was followed by additional IMF rescue packages in Turkey, Russia, Argentina, and Brazil. IMF loans increased again in late 2008 as the global financial crisis took hold. Between 2008 and 2010, the IMF made more than \$100 billion in loans to troubled economies such as Latvia, Greece, and Ireland. In April 2009, in response to the growing financial crisis, major IMF members agreed to triple the institution's resources from \$250 billion to \$750 billion, thereby giving the IMF the financial leverage to act aggressively in times of global financial crisis.

The IMF's activities have expanded because periodic financial crises have continued to hit many economies in the post-Bretton Woods era. The IMF has repeatedly lent money to nations experiencing financial crises, requesting in return that the governments enact certain macroeconomic policies. Critics of the IMF claim these policies have not always been as beneficial as the IMF might have hoped and, in some cases, may have made things worse. Following the IMF loans to several Asian economies, these criticisms reached new levels, and a vigorous debate was waged as to the appropriate role of the IMF. In this section, we discuss some of the main challenges the IMF has had to deal with over the past three decades and review the ongoing debate over the role of the IMF.

currency crisis

Occurs when a speculative attack on the exchange value of a currency results in a sharp depreciation in the value of the currency or forces authorities to expend large volumes of international currency reserves and sharply increase interest rates to defend the prevailing exchange rate.

banking crisis

A loss of confidence in the banking system that leads to a run-on banks, as individuals and companies withdraw their deposits.

foreign debt crisis

Situation in which a country cannot service its foreign debt obligations, whether private-sector or government debt.

FINANCIAL CRISES IN THE POST-BRETTON WOODS ERA

A number of broad types of financial crises have occurred over the past 30 years, many of which have required IMF involvement. A **currency crisis** occurs when a speculative attack on the exchange value of a currency results in a sharp depreciation in the value of the currency or forces authorities to expend large volumes of international currency reserves and sharply increase interest rates to defend the prevailing exchange rate. This happened in Pakistan in 2018–2019, and the IMF stepped in to help stabilize the value of the Pakistani currency, the rupee, on foreign exchange markets by lending it foreign currency (see the opening case). A **banking crisis** refers to a loss of confidence in the banking system that leads to a run on banks, as individuals and companies withdraw their deposits. This is what happened in Iceland in 2008. The experience of Iceland with the IMF is discussed in depth in the accompanying Country Focus. A **foreign debt crisis** is a situation in which a country cannot service its foreign debt obligations, whether private-sector or government debt. This happened to Greece, Ireland, and Portugal in 2010 and was also an issue in the most recent IMF intervention in Pakistan.

COUNTRY FOCUS

The IMF and Iceland's Economic Recovery

When the global financial crisis hit in 2008, tiny Iceland suffered more than most countries. The country's three biggest banks had been expanding at a breakneck pace since 2000, when the government privatized the banking sector. With a population of around 320,000, Iceland was too small for the banking sector's ambitions, so the banks started to expand into other Scandinavian countries and the United Kingdom. The banks entered local mortgage markets,

purchased foreign financial institutions, and opened foreign branches—attracting depositors by offering high interest rates. The expansion was financed by debt, much of it structured as short-term loans that had to be regularly refinanced. By early 2008, the three banks held debts that amounted to almost six times the value of the entire economy of Iceland! So long as they could periodically refinance this debt, it was not a problem. However, in 2008, global financial markets imploded following the bankruptcy of Lehman Brothers and the collapse of the U.S. housing market. In the aftermath, financial markets froze. The Icelandic banks found that they could not refinance their debt, and they faced bankruptcy.

The Icelandic government lacked the funds to bail out the banks, so it decided to let the big three fail. In quick succession, the local stock market plunged 90 percent, and unemployment increased ninefold. The krona, Iceland's currency, plunged on foreign exchange markets, pushing up the price of imports, and inflation soared to 18 percent. Iceland appeared to be in free fall. The economy shrank by almost 7 percent in 2009 and another 4 percent in 2010.

To stem the decline, the government secured \$10 billion in loans from the International Monetary Fund (IMF) and other countries. The Icelandic government stepped in to help local depositors, seizing the domestic assets of the Icelandic banks and using IMF and other loans to backstop deposit guarantees. Far from implementing austerity measures to solve the crisis, the Icelandic government looked for ways to shore up consumer spending. For example, the government provided means-tested subsidies to reduce the mortgage interest expenses of borrowers. The idea was to stop domestic consumer spending from tanking and further depressing the economy.

With the financial system stabilized, thanks to the IMF and other foreign loans, what happened next is an object lesson in the value of having a floating currency. The fall in the value of the krona helped boost Iceland's exports, such as fish and aluminum, while depressing demand for costly imports, such as automobiles. By 2009, the krona was worth half as much against the U.S. dollar and euro as it had been in 2007 before the crisis. Iceland's exports surged and imports slumped. While the high cost of imports did stoke inflation, booming exports started to pump money back into the Icelandic economy. In 2011, the economy grew again at a 3.1 percent annual rate. This was followed by 2.7 percent growth in 2012 and 4 percent growth in 2013, while unemployment fell from a high of nearly 10 percent to 4.4 percent at the end of 2013.

Sources: C. Forelle, "In European Crisis, Iceland Emerges as an Island of Recovery," *The Wall Street Journal*, May 19, 2012, pp. A1, A10; "Coming in from the Cold," *The Economist*, December 16, 2010; C. Duxbury, "Europe Gets Cold Shoulder in Iceland," *The Wall Street Journal*, April 26, 2012; "Iceland," *The World Factbook 2013* (Washington, DC: Central Intelligence Agency, 2013).

The various types of financial crises tend to have common underlying macroeconomic causes: high relative price inflation rates, a widening current account deficit, excessive expansion of domestic borrowing, high government deficits, and asset price inflation (such as sharp increases in stock and property prices).¹⁰ At times, elements of currency, banking, and debt crises may be present simultaneously, as in the 1997 Asian crisis, the 2000–2002 Argentine crisis, the 2010 crisis in Ireland, and Pakistan's 2018–2019 crisis.

To assess the frequency of financial crises, the IMF looked at the macroeconomic performance of a group of 53 countries from 1975 to 1997 (22 of these countries were developed nations, and 31 were developing countries).¹¹ The IMF found that 158 currency crises had occurred, including 55 episodes in which a country's currency declined by more than 25 percent. Also, 54 banking crises took place. The IMF's data suggest that developing nations were more than twice as likely as developed nations to experience currency and banking crises. We should not be surprised, therefore, that most of the IMF's loan activities since the mid-1970s have been targeted toward developing nations.

In 1997, several Asian currencies started to fall sharply as international investors came to realize that a speculative investment bubble had developed in the region. Those investors took their money out of local currencies, changing it into U.S. dollars, so the local currencies started to fall precipitously. The currency declines started in Thailand and then, in a process of contagion, quickly spread to other countries in the region. Stabilizing those currencies required massive help from the IMF. In the case of South Korea, local enterprises had built up huge debt loads as they invested heavily in new industrial capacity. By 1997, they found they had too much industrial capacity and could not generate the income required to service their debt. South Korean banks and companies had also made the mistake of borrowing in dollars, much of it in the form of short-term loans that would come due within a year. Thus, when the Korean won started to decline in the autumn of 1997 in sympathy with the problems elsewhere in Asia, South Korean companies saw their debt obligations balloon. Several large companies were forced to file for bankruptcy. This triggered a decline in the South Korean currency and stock market that was difficult to halt.

With its economy on the verge of collapse, the South Korean government requested \$20 billion in standby loans from the IMF on November 21. As the negotiations progressed, it became apparent that South Korea was going to need far more than \$20 billion. On December 3, 1997, the IMF and South Korean government reached a deal to lend \$55 billion to the country. The agreement with the IMF called for South Koreans to open their economy and banking system to foreign investors. South Korea also pledged to restrain Korea's largest enterprises, the *chaebol*, by reducing their share of bank financing and requiring them to



The IMF Managing Director Christine Lagarde addresses the Development Committee at the World Bank Headquarters in Washington, DC.

Stephen Jaffe/IMF/Handout/Getty Images News/Getty Images

publish consolidated financial statements and undergo annual independent external audits. On trade liberalization, the IMF said South Korea would comply with its commitments to the World Trade Organization to eliminate trade-related subsidies and restrictive import licensing and would streamline its import certification procedures, all of which should open the South Korean economy to greater foreign competition.¹²

EVALUATING THE IMF'S POLICY PRESCRIPTIONS

By 2019, the IMF had programs in around 40 countries that were struggling with economic and/or currency problems. All IMF loan packages come with conditions attached. Until very recently, the IMF has always insisted on a combination of tight macroeconomic policies, including cuts in public spending, higher interest rates, and tight monetary policy. It has also often pushed for the deregulation of sectors formerly protected from domestic and foreign competition, privatization of state-owned assets, and better financial reporting from the banking sector. These policies are designed to cool overheated economies by reining in inflation and reducing government spending and debt. This set of policy prescriptions has come in for tough criticisms from many observers, and the IMF itself has started to modify its approach.¹³

Inappropriate Policies

One criticism is that the IMF's traditional policy prescriptions represent a "one-size-fits-all" approach to macroeconomic policy that is inappropriate for many countries. In the case of the 1997 Asian crisis, critics argue that the tight macroeconomic policies imposed by the IMF were not well suited to countries that are suffering not from excessive government spending and inflation but from a private-sector debt crisis with deflationary undertones.¹⁴

In South Korea, for example, the government had been running a budget surplus for years (it was 4 percent of South Korea's GDP in 1994–1996), and inflation was low at about 5 percent. South Korea had the second-strongest financial position of any country in the Organisation for Economic Co-operation and Development. Despite this, critics say, the IMF insisted on applying the same policies that it applies to countries suffering from high inflation. The IMF required South Korea to maintain an inflation rate of 5 percent. However, given the collapse in the value of its currency and the subsequent rise in price for imports such as oil, critics claimed inflationary pressures would

inevitably increase in South Korea. So to hit a 5 percent inflation rate, the South Koreans would be forced to apply an unnecessarily tight monetary policy. Short-term interest rates in South Korea did jump from 12.5 to 21 percent immediately after the country signed its initial deal with the IMF. Increasing interest rates made it even more difficult for companies to service their already excessive short-term debt obligations, and critics used this as evidence to argue that the cure prescribed by the IMF might actually increase the probability of widespread corporate defaults, not reduce them.

At the time, the IMF rejected this criticism. According to the IMF, the central task was to rebuild confidence in the won. Once this was achieved, the won would recover from its oversold levels, reducing the size of South Korea's dollar-denominated debt burden when expressed in won and making it easier for companies to service their debt. The IMF also argued that by requiring South Korea to remove restrictions on foreign direct investment, it would encourage foreign capital to flow into the country to take advantage of cheap assets. This, too, would increase demand for the Korean currency and help improve the dollar/won exchange rate.

South Korea did recover fairly quickly from the crisis, supporting the position of the IMF. Although the economy contracted by 7 percent in 1998, by 2000, it had rebounded and grew at a 9 percent rate (measured by growth in GDP). Inflation, which peaked at 8 percent in 1998, fell to 2 percent by 2000, and unemployment fell from 7 to 4 percent over the same period. The won hit a low of \$1 = W1,812 in early 1998 but by 2000 was back to an exchange rate of around \$1 = W1,200, at which it seems to have stabilized.

Moral Hazard

A second criticism of the IMF is that its rescue efforts are exacerbating a problem known to economists as moral hazard. **Moral hazard** arises when people behave recklessly because they know they will be saved if things go wrong. Critics point out that many Japanese and Western banks were far too willing to lend large amounts of capital to overleveraged Asian companies during the boom years of the 1990s. These critics argued that the banks should have been forced to pay the price for their rash lending policies, even if that meant some banks would have closed.¹⁵ Only by taking such drastic action, the argument goes, will banks learn the error of their ways and not engage in rash lending in the future. By providing support to these countries, the IMF is reducing the probability of debt default and in effect bailing out the banks whose loans gave rise to this situation.

This argument ignores two critical points. First, if some Japanese or Western banks with heavy exposure to the troubled Asian economies were forced to write off their loans due to widespread debt default, the impact would have been difficult to contain. The failure of large Japanese banks, for example, could have triggered a meltdown in the Japanese financial markets. That would almost inevitably lead to a serious decline in stock markets around the world, which was the very risk the IMF was trying to avoid by stepping in with financial support. Second, implying that some banks have not had to pay the price for rash lending policies is incorrect. The IMF insisted on the closure of banks in South Korea, Thailand, and Indonesia after the 1997 Asian financial crisis. Foreign banks with short-term loans outstanding to South Korean enterprises have been forced by circumstances to reschedule those loans at interest rates that do not compensate for the extension of the loan maturity.

Lack of Accountability

The final criticism of the IMF is that it has become too powerful for an institution that lacks any real mechanism for accountability.¹⁶ The IMF has determined macroeconomic policies in those countries, yet according to critics such as noted economist Jeffrey Sachs, the IMF, with a staff of less than 1,000, lacks the expertise required to do a good job. Evidence of this, according to Sachs, can be found in the fact that the IMF was singing the praises of the Thai and South Korean governments only months before both countries lurched into crisis. Then the IMF put together a draconian program for South Korea without having deep knowledge of the country. Sachs's solution to this problem is to reform the IMF so it makes greater use of outside experts and its operations are open to greater outside scrutiny.

Observations

As with many debates about international economics, it is not clear which side is correct about the appropriateness of IMF policies. In some cases, one can argue that IMF policies had been counterproductive or had only limited success. For example, one might question the success

moral hazard

Arises when people behave recklessly because they know they will be saved if things go wrong.

of the IMF's involvement in Turkey given that the country has had to implement some 18 IMF programs since 1958! But the IMF can also point to some notable accomplishments, including its success in containing the Asian crisis, which could have rocked the global international monetary system to its core, and its actions in 2008–2010 to contain the global financial crisis, quickly stepping in to rescue Iceland, Ireland, Greece, and Latvia. Similarly, many observers give the IMF credit for its deft handling of politically difficult situations and for successfully promoting a free market philosophy.

Several years after the IMF's intervention, the economies of Asia recovered. Certainly, the kind of catastrophic implosion that might have occurred had the IMF not stepped in had been averted, and although some countries still faced considerable problems, it is not clear that the IMF should take much blame for this. The IMF cannot force countries to adopt the policies required to correct economic mismanagement. While a government may commit to taking corrective action in return for an IMF loan, internal political problems may make it difficult for a government to act on that commitment. In such cases, the IMF is caught between a rock and a hard place, because if it decided to withhold money, it might trigger financial collapse and the kind of contagion that it seeks to avoid.

Finally, it is notable that in recent years the IMF has started to change its policies. In response to the global financial crisis of 2008–2009, the IMF began to urge countries to adopt policies that included fiscal stimulus and monetary easing—the direct opposite of what the fund traditionally advocated. Some economists in the fund are also now arguing that higher inflation rates might be a good thing, if the consequence is greater growth in aggregate demand, which would help pull nations out of recessionary conditions. The IMF, in other words, is starting to display the very flexibility in policy responses that its critics claim it lacks. While the traditional policy of tight controls on fiscal policy and tight monetary policy targets might be appropriate for countries suffering from high inflation rates, the Asian economic crisis and the 2008–2009 global financial crisis were caused not by high inflation rates but by excessive debt, and the IMF's “new approach” seems tailored to deal with this.¹⁷



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360° View: Managerial Implications



LO11-6

Explain the implications of the global monetary system for management practice.

CURRENCY MANAGEMENT, BUSINESS STRATEGY, AND GOVERNMENT RELATIONS

The implications for international businesses of the material discussed in this chapter fall into three main areas: currency management, business strategy, and corporate–government relations.

Currency Management

An obvious implication with regard to currency management is that companies must recognize that the foreign exchange market does not work quite as depicted in Chapter 10. The current system is a mixed system in which a combination of government intervention and speculative activity can drive the foreign exchange market. Companies engaged in significant foreign exchange activities need to be aware of this and to adjust their foreign exchange transactions accordingly. For example, the currency management unit of Caterpillar claims it made millions of dollars in the hours following the announcement of the Plaza Accord by selling dollars and buying currencies that it expected to appreciate on the foreign exchange market following government intervention.

Under the present system, speculative buying and selling of currencies can create very volatile movements in exchange rates (as exhibited by the rise and fall of the dollar during the 1980s and the Asian currency crisis of the late 1990s). Contrary to the predictions of the purchasing power parity theory (see Chapter 10), exchange rate movements during the 1980s and 1990s often did not seem to be strongly influenced by relative inflation rates. Insofar as volatile exchange rates increase foreign exchange risk, this is not good news for business. On the other hand, as we saw in Chapter 10, the foreign exchange market has developed a number of instruments, such as the forward market and swaps, that can help ensure against foreign exchange risk. Not surprisingly, use of these instruments has increased markedly since the breakdown of the Bretton Woods system in 1973.

Business Strategy

The volatility of the current global exchange rate regime presents a conundrum for international businesses. Exchange rate movements are difficult to predict, and yet their movements can have a major impact on a business's competitive position. For a detailed example, see the accompanying Management Focus on Airbus. Faced with uncertainty about the future value of currencies, firms can utilize the forward exchange market, which Airbus has done. However, the forward exchange market is far from perfect as a predictor of future exchange rates (see Chapter 10). In addition, obtaining adequate insurance coverage for exchange rate changes that might occur several years in the future is difficult, if not impossible. The forward market tends to offer coverage for exchange rate changes a few months—not years—ahead. Given this problem, it makes sense to pursue strategies that will increase the company's strategic flexibility in the face of unpredictable exchange rate movements—that is, to pursue strategies that reduce the economic exposure of the firm (which we first discussed in Chapter 10).

MANAGEMENT FOCUS

Airbus and the Euro

Airbus had reason to celebrate in 2003; for the first time in the company's history, it delivered more commercial jet aircraft than long-time rival Boeing. Airbus delivered 305 planes in 2003, compared to Boeing's 281. The celebration, however, was muted because the strength of the euro against the U.S. dollar was casting a cloud over the company's future. Airbus, which is based in Toulouse, France, prices planes in dollars, just as Boeing has always done. But more than half of Airbus's costs are in euros. So as the dollar drops in value against the euro—and it dropped by more than 50 percent between 2002 and the end of 2009—Airbus's costs rise in proportion to its revenue, squeezing profits in the process.

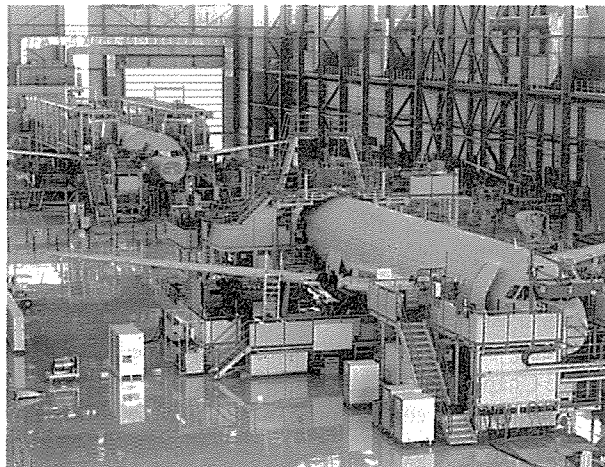
In the short run, the fall in the value of the dollar against the euro did not hurt Airbus. The company fully hedged its dollar exposure in 2005 and was mostly hedged for 2006. However, anticipating that the dollar would stay weak

against the euro, Airbus started to take other steps to reduce its economic exposure to a strong European currency. Recognizing that raising prices was not an option given the strong competition from Boeing, Airbus decided to focus on reducing its costs. As a step toward doing this, Airbus gave U.S. suppliers a greater share of work on new aircraft models, such as the A380 superjumbo and the A350. The company also shifted supply work on some of its older models from European to U.S.-based suppliers. This shift increased the proportion of its costs that were in dollars, making profits less vulnerable to a rise in the value of the euro and reducing the costs of building an aircraft when they were converted back into euros.

In addition, Airbus pushed its European-based suppliers to start pricing in U.S. dollars. Because the costs of many suppliers were in euros, the suppliers found that to comply with Airbus's wishes, they too had to move more work to the United States or to countries whose currency is pegged to the U.S. dollar. Thus, one large French-based supplier,

Zodiac, announced that it was considering acquisitions in the United States. Not only was Airbus pushing suppliers to price components for commercial jet aircraft in dollars, but the company was also requiring suppliers to its A400M program, a military aircraft that will be sold to European governments and priced in euros, to price components in U.S. dollars. Beyond these steps, the CEO of EADS, Airbus's parent company, publicly stated that it might be prepared to assemble aircraft in the United States if that would help win important U.S. contracts. While this strategy made good sense for years, it worked against Airbus between mid-2014 and 2018 as the dollar rose rapidly against the euro.

Sources: D. Michaels, "Airbus Deliveries Top Boeing's; but Several Obstacles Remain," *The Wall Street Journal*, January 16, 2004, p. A9; J. L. Gerondeau, "Airbus Eyes U.S. Suppliers as Euro Gains," *The Seattle Times*, February 21, 2004, p. C4; "Euro's Gains Create Worries in Europe," *HoustonChronicle.com*, January 13, 2004, 3; K. Done, "Soft Dollar and A380 Hitches Lead to EADS Losses," *Financial Times*, November 9, 2006, p. 32.



Aircraft factory assembly line.

Xenotar/E+/Getty Images

Maintaining strategic flexibility can take the form of dispersing production to different locations around the globe as a real hedge against currency fluctuations (this seems to be what Airbus has considered). Consider the case of Daimler-Benz, Germany's export-oriented automobile and aerospace company. In June 1995, the company stunned the German business community when it announced it expected to post a severe loss in 1995 of about \$720 million. The cause was Germany's strong currency, which had appreciated by 4 percent against a basket of major currencies since the beginning of 1995 and had risen by more than 30 percent against the U.S. dollar since late 1994. By mid-1995, the exchange rate against the dollar stood at $\$1 = \text{DM}1.38$. Daimler's management believed it could not make money with an exchange rate under $\$1 = \text{DM}1.60$. Daimler's senior managers concluded the appreciation of the mark against the dollar was probably permanent, so they decided to move substantial production outside of Germany and increase purchasing of foreign components. The idea was to reduce the vulnerability of the company to future exchange rate movements. Even before the company's acquisition of Chrysler Corporation in 1998, the Mercedes-Benz division planned to produce 10 percent of its cars outside Germany, mostly in the United States, by 2000. Similarly, the move by Japanese automobile companies to expand their productive capacity in the United States and Europe can be seen in the context of the increase in the value of the yen between 1985 and 1995, which raised the price of Japanese exports. For the Japanese companies, building production capacity overseas was a hedge against continued appreciation of the yen (as well as against trade barriers).

Another way of building strategic flexibility and reducing economic exposure involves contracting out manufacturing. This allows a company to shift suppliers from country to country in response to changes in relative costs brought about by exchange rate movements. However, this kind of strategy may work only for low-value-added manufacturing (e.g., textiles), in which the individual manufacturers have few if any firm-specific skills that contribute to the value of the product. It may be less appropriate for high-value-added manufacturing, in which firm-specific technology and skills add significant value to the product (e.g., the heavy equipment industry) and in which switching costs are correspondingly high. For high-value-added manufacturing, switching suppliers will lead to a reduction in the value that is added, which may offset any cost gains arising from exchange rate fluctuations.

The roles of the IMF and the World Bank in the current international monetary system also have implications for business strategy. Increasingly, the IMF has been acting as the macroeconomic police of the world economy, insisting that countries seeking significant borrowings adopt IMF-mandated macroeconomic policies. These policies typically include anti-inflationary monetary policies and reductions in government spending. In the short run, such policies usually result in a sharp contraction of demand. International businesses selling or producing in such countries need to be aware of this and plan accordingly. In the long run, the kind of policies imposed by the IMF can promote economic growth and an expansion of demand, which create opportunities for international business.

Corporate–Government Relations

As major players in the international trade and investment environment, businesses can influence government policy toward the international monetary system. For example, intense government lobbying by U.S. exporters helped convince the U.S. government that intervention in the foreign exchange market was necessary. With this in mind, business can and should use its influence to promote an international monetary system that facilitates the growth of international trade and investment. Whether a fixed or floating regime is optimal is a subject for debate. However, exchange rate volatility such as the world experienced during the 1980s and 1990s creates an environment less conducive to international trade and investment than one with more stable exchange rates. Therefore, it would seem to be in the interests of international business to promote an international monetary system that minimizes volatile exchange rate movements, particularly when those movements are unrelated to long-run economic fundamentals.

Key Terms

international monetary system, p. 333

floating exchange rate, p. 333

pegged exchange rate, p. 334

managed-float system, p. 334

dirty-float system, p. 334

fixed exchange rate, p. 334

European Monetary System (EMS), p. 334

dollarization, p. 334

gold standard, p. 335

gold par value, p. 336

balance-of-trade equilibrium, p. 336

currency board, p. 349

currency crisis, p. 350

banking crisis, p. 350

foreign debt crisis, p. 350

moral hazard, p. 353

Summary

This chapter explained the workings of the international monetary system and pointed out its implications for international business. The chapter made the following points:

1. The gold standard is a monetary standard that pegs currencies to gold and guarantees convertibility to gold. It was thought that the gold standard contained an automatic mechanism that contributed to the simultaneous achievement of a balance-of-payments equilibrium by all countries. The gold standard broke down during the 1930s as countries engaged in competitive devaluations.
2. The Bretton Woods system of fixed exchange rates was established in 1944. The U.S. dollar was the central currency of this system; the value of every other currency was pegged to its value. Significant exchange rate devaluations were allowed only with the permission of the International Monetary Fund (IMF). The role of the IMF was to maintain order in the international monetary system (a) to avoid a repetition of the competitive devaluations of the 1930s and (b) to control price inflation by imposing monetary discipline on countries.
3. The fixed exchange rate system collapsed in 1973, primarily due to speculative pressure on the dollar following a rise in U.S. inflation and a growing U.S. balance-of-trade deficit.
4. Since 1973, the world has operated with a floating exchange rate regime, and exchange rates have become more volatile and far less predictable. Volatile exchange rate movements have helped reopen the debate over the merits of fixed and floating systems.
5. The case for a floating exchange rate regime claims (a) such a system gives countries autonomy regarding their monetary policy and (b) floating exchange rates facilitate smooth adjustment of trade imbalances.
6. The case for a fixed exchange rate regime claims (a) the need to maintain a fixed exchange rate imposes monetary discipline on a country; (b) floating exchange rate regimes are vulnerable to speculative pressure; (c) the uncertainty that accompanies floating exchange rates dampens the growth of international trade and investment; and (d) far from correcting trade imbalances, depreciating a currency on the foreign exchange market tends to cause price inflation.
7. In today's international monetary system, some countries have adopted floating exchange rates; some have pegged their currency to another currency, such as the U.S. dollar; and some have pegged their currency to a basket of other currencies, allowing their currency to fluctuate within a zone around the basket.
8. In the post-Bretton Woods era, the IMF has continued to play an important role in helping countries navigate their way through financial crises by lending significant capital to embattled governments and by requiring them to adopt certain macroeconomic policies.

9. An important debate is occurring over the appropriateness of IMF-mandated macroeconomic policies. Critics charge that the IMF often imposes inappropriate conditions on developing nations that are the recipients of its loans.
10. The current managed-float system of exchange rate determination has increased the importance of currency management in international businesses.
11. The volatility of exchange rates under the current managed-float system creates both opportunities and threats. One way of responding to this volatility is for companies to build strategic flexibility and limit their economic exposure by dispersing production to different locations around the globe by contracting out manufacturing (in the case of low-value-added manufacturing) and other means.

Critical Thinking and Discussion Questions

1. Why did the gold standard collapse? Is there a case for returning to some type of gold standard? What is it?
2. What opportunities might current IMF lending policies to developing nations create for international businesses? What threats might they create?
3. Do you think the standard IMF policy prescriptions of tight monetary policy and reduced government spending are always appropriate for developing nations experiencing a currency crisis? How might the IMF change its approach? What would the implications be for international businesses?
4. Debate the relative merits of fixed and floating exchange rate regimes. From the perspective of an international business, what are the most important criteria in a choice between the systems? Which system is the more desirable for an international business?
5. Imagine that Canada, the United States, and Mexico decide to adopt a fixed exchange rate system. What would be the likely consequences of such a system for (a) international businesses and (b) the flow of trade and investment among the three countries?
6. Reread the Country Focus, “The IMF and Iceland’s Economic Recovery,” and then answer the following questions:
 - a. What were the main causes of Iceland’s economic troubles in 2008?
 - b. Was Iceland facing a classic currency crisis, or was this a banking crisis?
 - c. How did Iceland recover from its 2008–2009 crisis? What are the important lessons to draw from this case?
 - d. Iceland did not implement the austerity policies that are so often associated with IMF loans, and yet the economy recovered. Does this suggest that austerity policies do not work?
7. Reread the Country Focus, “China’s Exchange Rate Regime,” and then answer the following questions:
 - a. Why do you think that the Chinese historically pegged the value of the yuan to the U.S. dollar?
 - b. Why did the Chinese move to a managed-float system in 2005?
 - c. What are the benefits that China might gain by allowing the yuan to float freely against other major currencies such as the U.S. dollar and the euro? What are the risks? What do you think they should do?
 - d. Is there any evidence that the Chinese kept the level of their currency artificially low in the past to boost exports? Is China keeping it artificially low today?
 - e. What policy stance should the United States and the EU adopt toward China with regard to how it manages the value of its currency?

globalEDGE Research Task

Use the globalEDGE™ website (globaledge.msu.edu) to complete the following exercises:

1. The *Global Financial Stability Report* is a semiannual report published by the International Capital Markets division of the International Monetary Fund. The report includes an assessment of the risks facing the global financial markets. Locate and download the latest report to get an overview of the most important issues currently under discussion. Also, download a report from five years ago. How do issues from five years ago compare with financial issues identified in the current report?

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2. An important element to understanding the international monetary system is keeping updated on current growth trends worldwide. A German colleague told you yesterday that *Deutsche Bank Research* provides an effective way to stay informed on important topics in international finance from a European perspective. One area of focus for the site is emerging markets and economic and financial challenges faced by these markets. Find an emerging market research report for analysis. On which emerging market region did you choose to focus? What are the key takeaways from your chosen report?

CLOSING CASE



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The U.S. Dollar as the World's Reserve Currency

The U.S. dollar has been the world's reserve currency for the last 75 years. At the end of 2022, according to the IMF, 59.8 percent of all known foreign exchange reserves held by central banks around the world were U.S. dollars. The next most popular currency is the euro, which accounts for 19.7 percent of all foreign exchange reserves, followed by the Japanese yen (5.3 percent), British pound (4.6 percent), and Chinese yuan (2.8 percent). The dominance of the dollar makes it the *de facto* global reserve currency, even though it does not hold that title. Some 90 percent of all foreign exchange transactions involve U.S. dollars, 40 percent of the world's debt is issued in dollars, and the dollar is used in two-fifths of international payments. Most globally traded commodities, from oil to copper to sugar, are priced and traded in dollars, as are many industrial products, such as semiconductor chips. As of 2021, \$2,187 billion worth of dollar notes were in circulation, around half of which were circulating abroad. Many of these bills are in eastern Europe and Latin America, where they are often used as hard currency in day-to-day transactions. In Venezuela, for example, the U.S. dollar has supplanted the bolívar as the currency of choice because hyperinflation has made the bolívar all but worthless.

The dollar holds its unique position for a number of reasons. The 1944 Bretton Woods agreement gave the dollar a central position in the world's monetary system. Under that agreement, only the dollar could be converted into gold, while other countries pegged the value of their currency to that of the U.S. dollar. When President Nixon took the U.S. dollar off the gold standard in 1971, the dollar retained its position as the world's reserve currency due to the size of the U.S. economy, which was by far the largest in the world at that time. The U.S. economy continues to hold that position today, even though China has been catching up. Underpinning the continued strength of the dollar is the faith of foreigners in the ability of the U.S. Federal Reserve Board to limit inflation and maintain the value of the dollar and the ability of foreigners to move in and out of dollar-denominated assets with ease.

From time to time, naysayers have predicted the demise of the dollar as the world's reserve currency. In the early 2000s, some analysts thought the euro might increase in importance and ultimately overtake the dollar as the world's reserve currency, or at least match it. However, the European debt crisis that hit several eurozone countries in 2009, including Ireland, Greece, and Spain, damaged faith in the euro and demonstrated the difficulties associated with monetary union in the European Union. Since then, the share of the euro in global foreign exchange reserves has been stuck at around 20 percent.

More recently, some naysayers have pointed to the economic rise of China and argued that the Chinese yuan may replace the dollar as the world's reserve currency. After all, they note, China is already the world's second-largest economy, the largest

contributor to global growth, and the largest exporter. Within the next decade, it may also become the world's largest economy, so wouldn't it make sense for the yuan to replace the dollar as the world's reserve currency? The naysayers buttress their arguments by claiming that faith in the dollar is being undermined by U.S. fiscal and monetary policy. In 2020, the U.S. government embarked on a debt-driven spending spree, borrowing trillions of dollars, as the country grappled with the effects of COVID-19 on the U.S. economy. By 2022, this borrowing, coupled with expansion of the money supply by the Federal Reserve, helped to produce the highest inflation rate in America since the 1960s. This led to predictions that because inflation debases the value of the dollar, foreigners would lose faith in the currency as a stable store of value and decrease their dollar holdings. So far, however, there has been no sign of this happening, perhaps because the current inflation is a worldwide phenomenon, and the U.S. Federal Reserve has repeatedly signaled that it is committed to reducing dollar inflation through interest rate hikes.

While the Chinese would no doubt like the yuan to play a bigger role in international transactions and become a more important reserve currency, the currency remains a bit player in the global economy. For years, the argument was that China's financial markets lacked transparency, and regulations made it difficult for foreigners to issue, purchase, and trade yuan-denominated assets. Since 2015, the Chinese have reformed their financial markets, increasing transparency, access, and investor confidence, but foreigners still do not seem to want to hold on to yuan or use yuan in international transactions. What appears to be holding foreigners back is China's increasingly authoritarian political system under President Xi Jinping. Questions remain about whether China is truly committed to giving up state control over its currency and financial markets.

Two recent events illustrate these concerns. First, in 2020, the Chinese government blocked an initial public offering by Ant Group, the financial arm of the Chinese e-commerce giant Alibaba. Though the reasons for the government action are not publicly known, it seems likely that the Chinese government does not want a private enterprise to play too big a role in its financial system (some 88 percent of Chinese banks are state-controlled). Second, to protect the value of the yuan, in 2015 China imposed restrictions on capital leaving the country. Although these restrictions were focused on Chinese citizens and not foreigners, the imposition of capital controls signaled that when cross-border capital flows threaten the stability of the exchange rate and the control of the Chinese monetary authorities, the government will rein those flows in. These actions spooked foreign investors, who do not want to be limited in their ability to move in and out of yuan-denominated assets.

These events illustrate that, unlike U.S. financial markets, Chinese financial markets are still some way off from being free and open. As long as that is seen to be the case, foreigners are unlikely to want to hold too many yuan-denominated assets for fear that they might not be able to freely exchange them when needed. In short, absent significant financial reforms in China, the likelihood that the yuan will replace the dollar as the global reserve currency anytime soon seems improbable.

Sources: K. Amadeo, "Why the US Dollar Is the Global Currency," *The Balance*, July 23, 2020; D. McDowell, "Dollar Domsayers Are Wrong Again," *World Political Review*, January 19, 2021; "Global Trade's Dependence on Dollars Lessens Its Benefits," *The Economist*, August 29, 2020.

Case Discussion Questions

1. Why is the U.S. dollar the most commonly held foreign exchange reserve currency in the world?
2. What do you think has limited the appeal of (a) the euro and (b) the Chinese yuan as reserve currencies?
3. Under what conditions might the (a) euro and (b) the yuan increase their share of foreign exchange reserves?
4. What would have to occur for the dollar to lose its central role as the world's most widely held reserve currency?
5. If the U.S. dollar lost its role as the most popular reserve currency in the world, what would the implications be for the U.S. economy? For international businesses?

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Endnotes

1. The argument goes back to eighteenth-century philosopher David Hume. See D. Hume, "On the Balance of Trade," reprinted in *The Gold Standard in Theory and in History*, ed. B. Eichengreen (London: Methuen, 1985).
2. R. Solomon, *The International Monetary System, 1945-1981* (New York: Harper & Row, 1982).
3. International Monetary Fund, *World Economic Outlook, 2005* (Washington, DC: IMF, May 2005).
4. For an extended discussion of the dollar exchange rate in the 1980s, see B. D. Pauls, "US Exchange Rate Policy: Bretton Woods to the Present," *Federal Reserve Bulletin*, November 1990, pp. 891-908.
5. R. Miller, "Why the Dollar Is Giving Way," *BusinessWeek*, December 6, 2004, pp. 36-37.
6. For a feel for the issues contained in this debate, see P. Krugman, *Has the Adjustment Process Worked?*

- (Washington, DC: Institute for International Economics, 1991); "Time to Tether Currencies," *The Economist*, January 6, 1990, pp. 15–16; P. R. Krugman and M. Obstfeld, *International Economics: Theory and Policy* (New York: HarperCollins, 1994); J. Shelton, *Money Meltdown* (New York: Free Press, 1994); and S. Edwards, "Exchange Rates and the Political Economy of Macroeconomic Discipline," *American Economic Review* 86, no. 2 (May 1996), pp. 159–63.
7. The argument is made by several prominent economists, particularly Stanford University's Robert McKinnon. See R. McKinnon, "An International Standard for Monetary Stabilization," *Policy Analyses in International Economics* 8 (1984). The details of this argument are beyond the scope of this book. For a relatively accessible exposition, see P. Krugman, *The Age of Diminished Expectations* (Cambridge, MA: MIT Press, 1990).
 8. A. R. Ghosh and A. M. Gulde, "Does the Exchange Rate Regime Matter for Inflation and Growth?" *Economic Issues*, no. 2 (1997).
 9. "The ABC of Currency Boards," *The Economist*, November 1, 1997, p. 80.
 10. International Monetary Fund, *World Economic Outlook*, 1998 (Washington, DC: IMF, 1998).
 11. International Monetary Fund, *World Economic Outlook*, 1998.
 12. T. S. Shorrock, "Korea Starts Overhaul; IMF Aid Hits \$55 Billion," *Journal of Commerce*, December 8, 1997, p. 3A.
 13. See J. Sachs, "Economic Transition and Exchange Rate Regime," *American Economic Review* 86, no. 92 (May 1996), pp. 147–52; and J. Sachs, "Power unto Itself," *Financial Times*, December 11, 1997, p. 11.
 14. Sachs, "Power unto Itself."
 15. Martin Wolf, "Same Old IMF Medicine," *Financial Times*, December 9, 1997, p. 12.
 16. Sachs, "Power unto Itself."
 17. "New Fund, Old Fundamentals," *The Economist*, May 2, 2009, p. 78.